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A healthy and reliable rating profile expansion approach to address data sparsity in food recommendation systems

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Abstract
Food recommendation systems have become increasingly popular due to the proliferation of online food service websites. Accordingly, the ratings assigned by users are one of the most important resources in these systems. However, users generally express their opinions about a few foods, which results in data sparsity. Furthermore, food recommendation is a health-critical task, as recommending unhealthy foods to users may threaten their health. In this paper, we developed a novel rating profile expansion approach for food recommenders that considers both health and reliability measures. This approach enhances the efficiency of the user's rating profile by including healthy and reliable virtual ratings. Specifically, we introduce a probabilistic rating profile evaluation technique to determine whether a profile needs to be expanded. Then, those profiles with an insufficient number of ratings are automatically expanded by adding virtual ratings obtained using the opinions of users who belong to the target user's community. For this purpose, the users are grouped using a novel time-aware community detection algorithm based on their preferences. Moreover, a health-aware reliability measure is proposed so that only the most reliable virtual ratings are accounted for in the target user's rating profile expansion. Therefore, the developed approach not only mitigates issues stemming from sparse data in food recommendation systems but also makes them more effective in recommending healthy foods to users. Experiments conducted on two publicly available real-world datasets demonstrated that the developed system is superior to other baseline models.

Keywords Food recommendation system · Reliability · Health · Data sparsity · Community detection

1 Introduction

A vast amount of information has been generated on the Internet by the numerous online web services and applications. This has led to the well-known information overload problem, which limits the ability of Internet users to make decisions [1, 2]. To solve it, recommendation systems (RSs) that utilize information filtering techniques have been developed to assist users

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in identifying relevant records that best match their preferences [3–5]. To make personalized recommendations, RSs employ the historical data records of users and predict their future preferences [6, 7]. Because of their information filtering ability, RSs have been deployed in a broad range of real-world applications, such as information retrieval tools [8], e-learning [9], e-commerce [10], and social networks [11–14]. Overall, most developed RSs are categorized into collaborative filtering, content-based, and hybrid approaches [15, 16]. Collaborative filtering approaches make recommendations according to users' past behaviors and interests. To do this, they consider the opinions of their nearest neighbors about different items and select the most relevant ones as the recommendation list [17]. Content-based models generate the recommendation that best matches the user's preference based on the content of items [18]. For instance, genre, director, and actors, among other movie attributes, can be considered movie content in a movie recommendation system. In hybrid models, a combination of basic techniques (e.g., collaborative and content-based filtering) is used to construct more efficient models [19].

Consuming an unhealthy diet over an extended period is associated with high health risks, which lead to chronic diseases, such as diabetes, obesity, and various cancers [20]. This is a grave concern for both health authorities and the community; hence the problem of choosing appropriate healthy lifestyles and diets is of paramount importance [21]. Food RSs provide food recommendations to users. In recent decades, recommendation algorithms have garnered considerable attention for their role in assisting users in maintaining a healthy eating plan [22–25]. Although several food RSs have been developed [26–28], these systems still need to be improved due to critical unsolved shortfalls. First, unlike other RSs, such as movies, music, and book recommendations, the recommended foods critically affect health. Therefore, it is not enough for food RSs to use only the users' tastes in making food recommendations; the healthiness of the suggested foods should be considered. However, most previous works did not have any mechanism to handle such a critical issue, which restricted the analysis to the historical preferences of users [29]. Second, food RSs employ the user historical ratings as one of the data sources. However, this matrix is typically very sparse due to the reluctance of users to express their opinions on every food. This leads to a well-known problem called data sparsity, which makes recommendations difficult and degrades the RS performance considerably [30–32]. Third, food RSs encounter the user and item cold start problem [33]. User cold start refers to users whose rating profiles contain very few or no ratings for items. Item cold start is related to those items that have not received any ratings yet or have very few ratings. To deal with this problem, RSs expand the user's profile [34–37]. However, expanding a user's rating profile with unhealthy foods ultimately harms the user's health. Therefore, both health and reliability measures should be considered in designing a rating profile expansion technique for food RSs. Notably, the reliability concept in RSs has a different meaning in comparison to other applications where the reliability score is linked to the probability of failure [38, 39]. In the domain of RSs, the reliability concept is utilized to determine how much the predictions made by recommendation models can be trusted [40, 41]. Therefore, the accuracy of predictions is strongly correlated to their reliability values.

To address the data sparsity challenge in food RSs, we develop a novel rating profile expansion approach that simultaneously considers health aspects and reliability measures. This approach first evaluates the user's rating profile using a probabilistic model, indicating whether the ratings are sufficient to achieve reliable predictions. Thereafter, rating profiles lacking sufficient ratings are augmented by incorporating virtual ratings, sourced from the viewpoints of users within the community to which the target user pertains. The probabilistic model is a crucial part of the proposed rating profile expansion approach, as incorporating virtual ratings into profiles that already have sufficient ratings may lead to a significant

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decline in their efficiency. Profile expansion considers the reliability of the virtual ratings and the health content because incorporating unreliable virtual ratings into the target user's rating profile will lower its efficiency. Similarly, helping users shift toward a healthier diet should be the ultimate goal of the rating profile expansion approach. To achieve these two important purposes, we introduce a novel health-aware measure to assess the reliability of the calculated virtual ratings from both accuracy and health viewpoints. This criterion aids in assessing the credibility of a particular virtual rating for inclusion in the intended user's rating profile. Accordingly, the most reliable virtual ratings are selected for use in expanding the target user's rating profile. Finally, the unknown ratings are predicted using the expanded rating profiles. The novelties of the developed food RS method can be summarized as below:

- We develop a novel and effective rating profile expansion approach to overcome the data sparsity challenge in food recommendations, in which the user's rating profile is expanded by adding virtual ratings. According to our investigation, no previous work has focused on dealing with the data sparsity challenge in food RSs; thus, this paper is the first work in this context.
- We introduce a probabilistic model to specify whether a user's rating profile is to be expanded. Specifically, the model determines the required number of ratings for each rating profile to make accurate predictions.
- A novel time-aware community detection approach is introduced to group users according to their food preferences while considering the time to manage the shifts in users' eating habits over time. This algorithm incorporates preferences from users within the same community as the target user to calculate virtual ratings. This leads to two important achievements. First, as the users of the same community may have potentially similar tastes, the use of community members' ratings in the calculation of the virtual ratings for the target user yields more accurate predictions. Second, we only calculate the virtual ratings of those foods that have been rated by the users who belong to the community of the target user. This lessens the computational intricacy of the proposed approach since we refrain from computing virtual ratings for all food items.
- We introduce a health-aware reliability measure to assess the reliability of the calculated virtual ratings. This measure helps to ensure that the virtual ratings are healthy and reliable enough.
- We designed different experiments on two food RS datasets, and the results prove that the developed approach greatly outperforms the other food recommenders based on both accuracy and health-s core metrics.

The remainder of this paper is structured as follows. In Sect. 2, we delve into relevant literature. Section 3 outlines the specifics of our recommendation method. Following that, Sect. 4 delineates the experiments conducted and elaborates on the results. Conclusively, Sect. 5 wraps up the paper.

2 Related works

Many food RSs have been developed in recent years to help users discover new food products that match their interests while promoting a healthy diet. An emerging research field for personalized nutrition is providing tailored food intake advice based on an individual's physical, physiological, and personal data. In [42], the authors introduced a daily meal plan recommendation framework based on the combination of food content and user historical ratings. Specifically, a multi-criteria decision analysis technique was utilized to filter out foods that were not relevant to the target user's characteristics. Critically, however, they overlooked

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food ingredient information, which is crucial for understanding food content. Therefore, the authors of [43] formulated the food recommendation problem as user preference prediction by employing user historical data, food ingredients, and food images. Accordingly, the authors proposed a food RS called Hierarchical Attention-Based Food Recommendation (HAFR), which uses collaborative filtering to identify similar users to the target user, whereas the foods' ingredients and images were utilized to learn the preferences of users. To explore ingredient–ingredient and ingredient–recipe representations in greater depth, a graph convolutional network was suggested in [44]. The aim was to design a food RS that learns food contents, user preferences, and food relationships effectively. To this end, a propagation strategy was proposed in the graph convolutional network to make high-order representations. Despite the accuracy improvements in these works, their final performance may have been limited due to the constraints of the data resources they utilized for food recommendations. Consequently, researchers have endeavored to employ diverse data resources in certain food RSs. Mathur et al. [45] demonstrated the feasibility of using review information instead of explicit ratings in food RSs by proposing a graph theoretic-based approach, where a graph mining technique was utilized to calculate the similarity scores between foods and identify popular foods.

Most previous RSs cannot ensure that the suggested food options are beneficial for users' health. Recent research has shown that a considerable number of available foods in online food services are unhealthy, and users mainly face the challenge of finding healthier foods due to the increasing growth of food varieties [46]. To provide users with healthy food recommendations, some food RSs have been developed. Musto et al. [27] proposed a natural language justification that exploits the nutritional content, health risks, and suitability of food recommendations to encourage users toward healthier diets. To this end, the user characteristics and the food features were utilized to generate natural language justification as output in a way that helps users distinguish between healthy and unhealthy food recommendations. In [47], the authors designed a food recommendation model named Market2Dish by considering the health factors of recommended foods. This model uses food ingredients to select food candidates among numerous available foods. The textual health-related information of users was then employed to capture healthy foods from the candidates and make healthy food recommendations to users. The authors of [28] demonstrated that using only the preferences of users is not enough to make healthy food recommendations. Therefore, they utilized users' health-related characteristics in designing a health-aware recommender system to help users adopt healthier eating habits. Moreover, a holistic user model was introduced to re-rank the initial recommendations generated by the recommender system and quantify the extent to which the recommended foods accommodate users' preferences.

Data sparsity and cold start are widespread challenges in RSs, which negatively affect the quality of generated recommendations. Researchers have conducted research to address these issues. Singh et al. [48] conducted a study on alleviating data sparsity in RSs, in which the Bhattacharya coefficient was utilized as an effective similarity function for users whose common items are very few, or no co-rated item exists between them. In addition, they employed the content information of items alongside user ratings to generate more accurate recommendations. While the focus was on alleviating data sparsity, cold start received comparatively less attention. To tackle the challenges of data sparsity and cold start in context-aware RSs, auxiliary information, such as location data, was utilized in [49] to enhance recommendation accuracy, especially for users with insufficient available data. In particular, they devised a density-based clustering model to tackle the problem of sparse data, alongside developing a hybrid similarity function leveraging auxiliary information to identify the nearest neighbors of cold start users. In another work [33], a linked open data model was used to design a

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collaborative filtering-based RS that could address the data sparsity and the cold start problems simultaneously. Moreover, the proposed linked open data model employed the matrix factorization technique to better perform the recommendation process for cold start users.

Although most recommendation systems concentrate on enhancing the accuracy of predictions, the reliability of recommendations may also substantially impact their performance. For instance, in [41], the authors demonstrated a considerable correlation between the accuracy of predictions and their reliability values, where a prediction with a high reliability score has a lower probability of yielding the wrong recommendation. A limitation of this study was the oversight in evaluating how introducing the combination of prediction and reliability into user recommendations could have more accurately gauged the user's confidence in the RS over time. To this end, Bobadilla et al. [40] introduced two effective reliability measures to assess both prediction quality and recommendation quality. Fundamentally, these measures assume that predictions that have high reliability scores lead to more reliable recommendations as well. These proposed measures can be used in designing a wide range of RSs, including collaborative filtering-based, content-based, context-aware, and matrix factorization-based models. The proposed system demonstrates significant gains in rating prediction accuracy with minimal performance costs, whereas the incorporation of confidence levels increases performance in social networks with established user relationships as well as those without. To address this, a matrix factorization-based RS was proposed based on sentiment analysis and users' reviews [50]. To this end, a star-based dictionary construction model was developed to calculate the sentiment scores. Then, the reliability of these scores was evaluated using an effective measure to boost the efficiency of the developed RS.

The above-mentioned reveals that no previous work has focused on the data sparsity problem in food RSs as a whole. Real-world online food services suffer from this problem mainly due to the lack of sufficient ratings assigned by users to various foods. Moreover, the previous food recommendation models barely account for the reliability of predictions in their recommendation scheme. However, as we discussed earlier, incorporating reliability evaluation into food recommendations can guarantee healthy and reliable food recommendations to users. Unlike the previous works, our food recommendation model addresses these shortcomings using a novel user rating profile expansion approach.

3 Proposed method

In this section, we discuss the proposed food recommendation system called HRRPE, which is based on a novel Healthy and Reliable Rating Profile Expansion approach. The general structure of the proposed HRRPE method is based on six main steps: (1) rating profile evaluation, (2) community detection, (3) virtual rating prediction, (4) reliability evaluation, (5) rating profile expansion, and (6) recommendation. In the rating profile evaluation, we evaluate the quality of each rating profile using a probabilistic model, which determines whether the ratings for predicting accurate recommendations are sufficient. A time-aware community detection algorithm was developed in the second step to assign users to meaningful communities based on their food interests. Then, the obtained communities were used in the third step to predict virtual ratings, in which these ratings were calculated using the opinions of users within the community that the target user also belongs to. In the fourth step, a reliability assessment was performed for the predicted virtual ratings using a health-aware reliability measure, which considers both the health and accuracy factors simultaneously. Then, in the

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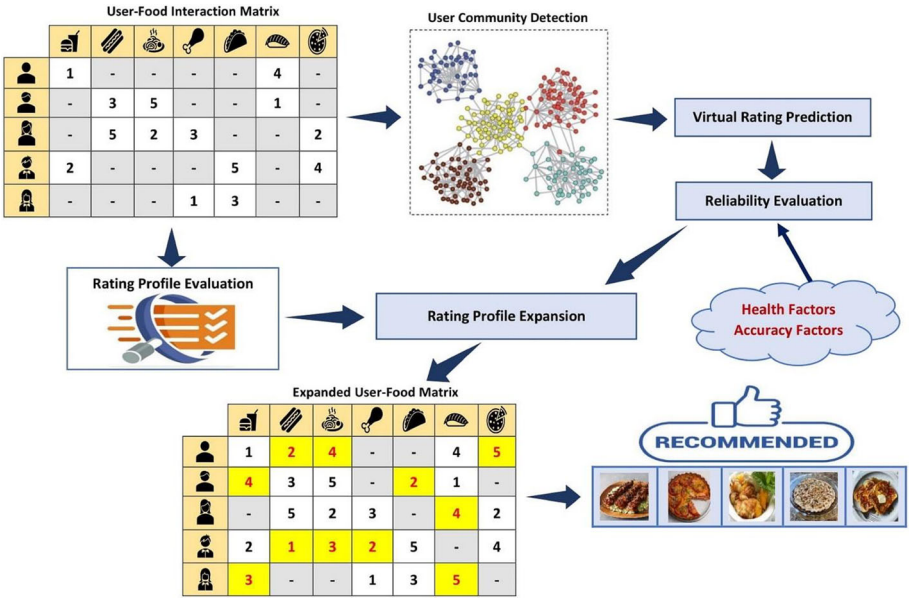


Fig. 1 Overview of the proposed HRRPE method. The user-food interaction matrix serves as the primary input data source, comprising the ratings allocated to foods by users. HRRPE includes six steps: (1) rating profile evaluation step, in which a probabilistic model is used to determine the minimum number of ratings needed to make reliable predictions, (2) User community detection step for detecting user communities using a graph-based community detection approach, (3) virtual rating prediction step to predict virtual ratings based on users' communities detected by the proposed community detection method, (4) reliability evaluation step, in which a new health-aware reliability measure is introduced to concurrently take health and accuracy aspects into account to evaluate the predicted virtual ratings, (5) rating profile expansion step to expand the users' rating profiles using a mechanism that adds some virtual ratings to the profiles with insufficient ratings to make accurate predictions; and (6) recommendation step, in which a prediction procedure is performed based on the expanded profiles to generate a list of food recommendations

rating profile expansion step, the most reliable virtual ratings were appended to the target rating profile. Finally, the unknown ratings were predicted according to the expanded profiles, and a list of relevant food recommendations was generated for the target user. The HRRPE's architecture is depicted in Fig. 1.

3.1 Problem definition

Each food recommendation system consists of users set and foods set. In this paper, we denote users set as $U = \{u_1, u_2, \ldots, u_{|U|}\}$. Moreover, foods set is denoted as $I = \{f_1, f_2, \ldots, f_{|I|}\}$. Users make ratings about available foods, thus forming a matrix named as a user–food interaction matrix, denoted by $R = \{r_{u,i}\} \in \mathbb{R}^{|U| \times |I|}$, where $r_{u,i}$ is the user u's rating assigned to food f_i. Assume that the ratings of the user–food interaction matrix are within the [minR, maxR] range, where minR and maxR refer to the lowest interest and highest interest, respectively. Most of the entities of the user–food interaction matrix are empty, implying that the users have not assigned any ratings to those foods. These empty entities are named unknown ratings. The main purpose of a food recommendation system is

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Table 1 A list of the main notations used in this study, along with their brief descriptions

Notation	Description
U	Users set
I	Foods set
R	User-food interaction matrix
$r_{u,i}$	User u's rating assigned to food f_i
minR	The lowest rating in R
maxR	The highest rating in R
n'_u	Minimum number of required ratings
δ	Uncertainty level of predictions
$G = (U, E, W)$	Graph of user-food relationships
E	Set of edges in graph G
$W \in \mathbb{R}^{\lvert U\rvert \times \lvert U\rvert}$	Weights matrix for graph G
λ	Time weight parameter
$\mathrm{VR}_{u,i}$	Virtual rating of food f_i for user u
$\mathrm{HU}(v)$	Health score for user v
$\mathrm{HF}(i)$	Health score for food f_i
$\mathrm{Rel}_{u,i}$	Health-aware reliability value
$K_{u,i}$	Nearest neighbor set for user u
N	Number of nearest neighbors
$P_{u,i}$	Predicted rating for user u on food f_i

to predict the missing ratings in the user–food interaction matrix using the historical preferences of users. Therefore, the problem is identifying to which extent the future rating of a user agrees with the available user–food rating matrix. As the output of a food recommendation system, a list of foods with the highest predicted ratings is provided to the target user. The main notations used in this paper along with their descriptions are listed in Table 1.

3.2 Rating profile evaluation

For food RSs to provide accurate recommendations, they necessitate an ample number of ratings, as their capability to predict unseen foods relies heavily on the quantity of ratings within each user's rating profile. Typically, the accuracy of the predicted ratings may degrade if there are not enough historical ratings in the rating profiles. In this stage of the HRRPE method, we devise a technique to assess the efficacy of historical ratings for each user within the user-food rating matrix. Accordingly, a probabilistic model is used to determine the minimum number of ratings needed for users to make reliable predictions. In this model, the number of required ratings is compared with the number of ratings provided by the target user. Therefore, the profiles whose number of ratings exceeds the calculated minimum can be deemed reliable. Conversely, if they dip beneath that threshold, they ought to be deemed as unreliable profiles. To assess the rating profile of user u, we initially compute the probability of user u assigning a rating r from the range [minR, maxR] [51]:

$$\alpha_{u,r} = \Pr\left[r_{u,i} = r\right] = \frac{n_{u,r}}{|I_u|} \tag{1}$$

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where the rating of food f_i provided by user u is denoted by $r_{u,i}$, $n_{u,r}$ is the number of foods rated by user u in the rating level r, and all ratings of user u are in the set I_u. Based on the majority rule, the true rating provided by the user u is calculated as follows:

$$l_u = \max_{r \in [\text{minR}, \text{maxR}]} \{\alpha_{u,r}\} \tag{2}$$

where the true rating is indicated by l_u for user u.

The model defines the minimum number of ratings required for each user to ensure that the predicted rating converges to the true rating with high probability. Consequently, the minimum requisite number of ratings for user u to yield dependable predictions is denoted by n'_u, as stated in Proposition 1.

Proposition 1 *The minimum number of user ratings needed to ensure a high probability that the predicted rating ($\widehat{l_u}$) will approach the actual rating (l_u) is:*

$$n'_u = \left(2\left(\alpha_{u,l_u} + \widetilde{\alpha}_u\right)\left(\alpha_{u,l_u} - \widetilde{\alpha}_u\right)^{-2} - 2 + \frac{4\left(\alpha_{u,l_u} - \widetilde{\alpha}_u\right)^{-1}}{3} \right) \times ln(maxR - minR)\delta^{-1} \tag{3}$$

where δ refers to the level of uncertainty of predictions and the second-highest value of $\alpha_{u,r}$, represented as $\widetilde{\alpha}_u$, is calculated as follows:

$$\widetilde{\alpha}_u = \max_{r \in [minR, maxR]} \{\alpha_{u,r} | r \neq l_u\} \tag{4}$$

Proof First, we use the following lemma.

Lemma 1 *The probability mass function (pmf) of the rating $r_{u,i}$ can be represented as $Pr[r_{u,i} = r] = \alpha_{u,r}$, where $r \in [minR, maxR]$, $u \in U$, and $i \in I_u$.*

Proof of Lemma 1 Consider $\rho = \left(\rho_{u,\text{minR}}, \ldots, \rho_{u,\text{maxR}}\right)$ as a function of Dirichlet distribution to model the probability distribution of ratings assigned by user u. Suppose ρ is a random sample obtained from Dirichlet($\alpha_{u,r}$). Because the conditional pmf of $r_{u,i}$ is described as $\Pr[r_{u,i} = r|\rho] = \rho_{u,r}$, this conditional pmf possesses a Dirichlet distribution fundamental property $\Pr[r_{u,i} = r] = \int \Pr[\rho]\Pr[r_{u,i} = r|\rho]d\rho = \int \Pr[\rho]\rho_{u,r}d\rho = \alpha_{u,r}$ [52].

Next, we propose the following theorem:

Theorem 1 [53] *Let $X = X_1 + X_2 + \cdots + X_n$ and $\sigma^2 = Var[X] = \sum_{i=1}^{n} Var[X_i]$, where the values of $X_1, X_2, \ldots, X_n$ are selected in the range $[0, 1]$ and they are considered as n independent random variables. Therefore, for each $\tau \geq 0$, the following two statements hold:*

$$\Pr[X \geq E[X] + \tau] < exp\left(-\frac{\tau^2}{2(\sigma^2 + \tau/3)}\right), \text{ and}$$

$$\Pr[X \leq E[X] - \tau] < exp\left(-\frac{\tau^2}{2(\sigma^2 + \tau/3)}\right).$$

From Eq. (3), it can be proved that if $|I_u| \geq n'_u$, then $\Pr[\widehat{l_u} = l_u] \geq 1 - \delta$. This outcome suggests that when the count of ratings in user u's profile exceeds n'_u, we can anticipate a

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precise prediction with a probability higher than $1-\delta$. Equation (3) is proven by applying Theorem 1 to demonstrate that $\Pr[\widehat{l_u} \neq l_u] \leq \delta$. In light of the fundamental probability arguments, we can derive the following equation:

$$\Pr[\widehat{l_u} \neq l_u] = \Pr\left[\bigcup_{r \neq l_u} \{\widehat{l_u} = r\}\right] \leq \sum_{r \neq l_u} \Pr[\widehat{l_u} = r] \leq \sum_{r \neq l_u} \Pr[n_{u,r} \geq n_{u,l_u}] \tag{5}$$

where l_u is derived from Eq. (2) and refers to the most frequent rating compared to all the ratings by user u. As previously stated, l_u is determined based on the majority rule.

We consider $\widehat{l_u}$ as the rating predicted by the food RS for the true rating l_u. Noticeably, if $\widehat{l_u} = l_u$ then, it can be inferred that the food RS achieves a correct prediction for l_u. The aim of Eq. (5) is to calculate $\Pr[\widehat{l_u} \neq l_u]$, representing the probability of obtaining an inaccurate rating for l_u through the food RS. In other words, if $\widehat{l_u} \neq l_u$ then, the initial assumption about l_u is false, which means that there is another rating $r \in [\text{minR}, \text{maxR}]$ and $n_{u,r} \geq n_{u,l_u}$. Meanwhile, the summation of $\Pr[n_{u,r} \geq n_{u,l_u}]$ for all $r \neq l_u$ is assumed as the overall probability value for predicting a false rating by the food RS. Let $R^r_{u,i}$ indicate a set of random variables defined as follows:

$$R^r_{u,i} = \begin{cases} 1, & \text{with probability } \Pr[r_{u,i} = r] \\ 0, & \text{with probability } \Pr[r_{u,i} = l_u] \\ 1/2, & \text{otherwise} \end{cases} \tag{6}$$

where $r \in [\text{minR}, \text{maxR}]$, $u \in U$, $i \in I_u$, and the pmf of $r_{u,i}$ is deduced from Lemma 1. If $R^r_u = \sum_{i \in I_u} R^r_{u,i}$, then we have $R^r_u = n_{u,r} + (|I_u| - n_{u,r} - n_{u,l_u})/2$. From this assumption, we derive the following equation:

$$R^r_u \geq \frac{|I_u|}{2} \Leftrightarrow n_{u,r} + \frac{|I_u| - n_{u,r} - n_{u,l_u}}{2} \geq \frac{|I_u|}{2} \Leftrightarrow n_{u,r} \geq n_{u,l_u} \tag{7}$$

Then, we derive the following equation from Eq. (7):

$$\Pr[n_{u,r} \geq n_{u,l_u}] = \Pr\left[R^r_u \geq \frac{|I_u|}{2}\right] \tag{8}$$

Drawing upon Eq. (3), for the target user u, if the number of ratings in user u's profile exceeds n'_u (i.e., $|I_u| \geq n'_u$), then the probability of attaining an accurate prediction (i.e., $\widehat{l_u} = l_u$) through the food RS will surpass $1-\delta$ (i.e., $\Pr[\widehat{l_u} = l_u] \geq 1-\delta$). As a result, Eq. (3) can be proved by demonstrating that $\Pr[\widehat{l_u} \neq l_u] \leq \delta$. To do this, we use Eqs. (5) and (8) to derive $\Pr[\widehat{l_u} \neq l_u] \leq \sum_{r \neq l_u} \Pr\left[H^r_u \geq \frac{|I_u|}{2}\right]$. Furthermore, the expectation and variance of H^r_u are defined in Eqs. (9) and (10), respectively, as follows:

$$E[H^r_u] = \frac{|I_u|}{2} - \frac{(\alpha_{u,l_u} - \alpha_{u,r})|I_u|}{2} \tag{9}$$

$$\text{Var}[H^r_u] = \frac{\left(\alpha_{u,l_u} + \alpha_{u,r} - (\alpha_{u,l_u} - \alpha_{u,r})^2\right)|I_u|}{4} \tag{10}$$

As we mentioned earlier, each rating in the food RS can be in the range $r \in [\text{minR}, \text{maxR}]$. As a result, it is essential to demonstrate that $\Pr\left[H^r_u \geq \frac{|I_u|}{2}\right] \leq \frac{\delta}{\text{maxR}-\text{minR}}, \forall r \neq l_u$ for

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provingPr$\left[\widehat{l_u} \neq l_u\right] \leq \delta$. Thus, Theorem 1 is applied to $\tau = \frac{(\alpha_{u,l_u} - \alpha_{u,r})|I_u|}{2}$ to finalize this proof as follows:

$$\begin{aligned}
\Pr\left[R_u^r \geq \frac{|I_u|}{2}\right] &= \Pr\left[R_u^r \geq E\left[R_u^r\right] + \tau\right] \leq \exp\left(-\frac{\tau^2}{2\left(\mathrm{Var}\left[R_u^r\right] + \frac{\tau}{3}\right)}\right) \\
&= \exp\left(-|I_u| \times \left(\frac{2\left(\alpha_{u,l_u} + \alpha_{u,r}\right)}{\left(\alpha_{u,l_u} - \alpha_{u,r}\right)^2} - 2 + \frac{4}{3\left(\alpha_{u,l_u} - \alpha_{u,r}\right)}\right)^{-1}\right) \\
&\leq \exp\left(-|I_u| \times \left(\frac{2\left(\alpha_{u,l_u} + \tilde{\alpha}_u\right)}{\left(\alpha_{u,l_u} - \tilde{\alpha}_u\right)^2} - 2 + \frac{4}{3\left(\alpha_{u,l_u} - \tilde{\alpha}_u\right)}\right)^{-1}\right) \leq \frac{\delta}{(\mathrm{maxR} - \mathrm{minR})}
\end{aligned} \tag{11}$$

Finally, the term $\Pr\left[H_u^r \geq \frac{|I_u|}{2}\right] \leq \frac{\delta}{\mathrm{maxR-minR}}, \forall r \neq l_u$ can be achieved from Eq. (11), resulting in $\Pr\left[\widehat{l_u} \neq l_u\right] \leq \delta$. Therefore, the correctness of Eq. (3) is proved theoretically, which means that if $|I_u| \geq n'_u$, then $\Pr\left[\widehat{l_u} = l_u\right] \geq 1 - \delta$. Through this probabilistic model, we can conclude that if $|I_u| \geq n'_u$, then an accurate rating can be predicted by the food RS with a probability higher than $1 - \delta$. Notably, parameter δ has a significant impact on how effectively rating profiles are evaluated. The lower value of this parameter means a higher probability of making accurate predictions and vice versa. The main purpose of the proposed rating profile evaluation strategy is to specify whether each rating profile contains sufficient explicit ratings. Accordingly, we first utilized Eq. (3) to compute n'_u for each user rating profile according to the initialized level of uncertainty (i.e., δ). Then, we compare the minimum with the number of ratings that have already been provided by the target user to determine whether this profile needs to be expanded. In this regard, for each target user u, the rating profile will be expanded only when $|I_u| < n'_u$; otherwise, the available real ratings without any changes are utilized in the recommendation process.

3.3 Community detection

In social network analysis, identifying similar communities of users is essential. In general, a community detection algorithm identifies clusters with low intercommunity similarity but high intracommunity similarity. It has been demonstrated that users belonging to a community exhibit significantly higher pairwise similarity scores, in stark contrast to users outside the community who exhibit markedly lower pairwise similarity scores. This step aims to develop a novel algorithm for detecting user communities. Accordingly, we map the users' space to a graph $G = (U, E, W)$, where U is users set, E represents the relationships between the users as the set of edges, and $W \in \mathbb{R}^{|U| \times |U|}$ refers to a weight matrix for the edges and is generated based on a time-aware similarity measure. Each user indicates a node in the graph G. Our developed time-aware similarity measure is defined as follows:

$$w_{u,v} = \frac{\sum_{i \in I_{u,v}} ((r_{u,i} - \bar{r}_u) \times (r_{v,i} - \bar{r}_v) \times \mathrm{TW}_{u,v,i})}{\sqrt{\sum_{i \in I_{u,v}} ((r_{u,i} - \bar{r}_u)^2 \times \mathrm{TW}_{u,v,i})} \sqrt{\sum_{i \in I_{u,v}} ((r_{v,i} - \bar{r}_v)^2 \times \mathrm{TW}_{u,v,i})}} \tag{12}$$

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where a set of food items rated by both users u and v is represented by $I_{u,v}$. The rating assigned to a food item f_i by user u is represented by $r_{u,i}$, while $\overline{r}_u$ is the average of all ratings assigned by user u. Additionally, $\mathrm{TW}_{u,v,i}$ denotes a time weight which is defined as follows:

$$\mathrm{TW}_{u,v,i} = \sqrt{e^{-\lambda(\mathrm{TL}-t_{u,i})} \times e^{-\lambda(\mathrm{TL}-t_{v,i})}} \tag{13}$$

where the rating's timestamp is denoted by $t_{u,i}$ and the maximum value of all available timestamps is indicated by TL. Moreover, we use a preset parameter named λ to control the influence of the time factor in calculating similarity values.

We obtain the users' communities based on the constructed graph $G = (U, E, W)$ using an effective algorithm. In this regard, the core nodes with a high amount of influence in the graph should be chosen from denser parts, as community detection algorithms rely heavily on central nodes. According to network structure, nodes with high node centrality can penetrate their neighbors. In a community with numerous connections, central nodes have substantial influence on group consistency, while bordering nodes act as mediators. Thus, we use a new linear centrality to give a higher score to the nodes that belong to the core of the community and whose influence value is high. To assess the presence of a central node in a community, the core dominance measure can be utilized in the following manner: [54]:

$$\mathrm{CoreDominance}(u) = \sum_{v \in U, v \neq u} w_{u,v} \tag{14}$$

where $w_{u,v}$ is calculated using Eq. (12).

The initial core nodes are related to the users with the highest core dominance values. To expand the core nodes, the algorithm adds all of the first-degree neighbors of the initial core nodes. The first-degree neighbors of the added nodes are then selected as expansion candidates. This process is repeated until the desired number of nodes is reached or there are no more expansion candidates available. Then, we use a membership-degree function to expand the central nodes. A strong community c refers to a community in which the sum of the weights between each node $u \in c$ and other nodes of c (i.e., A) is greater than the sum of the weights between node u and the nodes of other communities (i.e., B). It is proved that if $A \geq B$, then $A/(A + B) \geq 0.5$ [55]. As a result, the membership degree between node u and community c is defined as follows:

$$\mathrm{MembershipDegree}(u, c) = A/(A + B) \tag{15}$$

The membership-degree function is used to determine which nodes to include in the initial subset. A node will be added to the subset if its membership degree is greater than 0.5. This threshold is based on the idea of a strong community, where the ratio of internal edges within the community to the total sum of edges within and outside of the community's nodes should be more than 0.5. Moreover, when the nodes are weighed according to their similarity, each edge's weight is less than 1. As a result, the expanding step is rational when 0.5 is used as the threshold value of the membership degree in a weighted graph. Notably, choosing higher values for this threshold results in adding too many nodes in the expansion step, causing a significant decline in the proposed algorithm's performance. A node will be added to the initial subset if its membership degree is higher than the threshold value of 0.5. Additionally, the node's neighbors are considered for expansion. The process is iterated until no further nodes can be expanded, which results in detection of the first community. Then, we remove the selected nodes from the node list and iterate this process until all communities have been identified.

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Finally, outlier nodes that are not members of any community are identified. These outlier nodes will be merged with other communities. Accordingly, for each detected community, we calculate the embeddedness value of its users. The embeddedness criterion measures the proportion of a node u's neighbors that belongs to its own community. Therefore, this criterion can be calculated for each node u based on the internal degree $k_{\text{int}}(u)$ and the total degree $k_t(u)$ as follows [56]:

$$\text{Embeddedness}(u) = k_{\text{int}}(u)/k_t(u) \tag{16}$$

where Embeddedness(u) can be in the range of 0 when all neighbors are from different communities (i.e., $k_{\text{int}} = 0$) and 1 when all of the neighbors are from the same community (i.e., $k_{\text{int}} = k$). A user with the highest embeddedness value is considered the representative node of the community. Then, each outlier node will be merged into the community such that the weight between its representative node and the outlier node is the highest value compared to other detected communities. In this manner, the merging process continues until all outlier nodes are merged. Accordingly, trivial communities are merged, and detected communities will be of higher quality. The proposed community detection algorithm is described in the pseudo-code of Algorithm 1.

Algorithm 1 The proposed community detection algorithm

Inputs: User-food interaction matrix.
Output: Detected communities $\boldsymbol{C}$.
Algorithm:
Map the users' space to a graph $\boldsymbol{G} = (\boldsymbol{U}, \boldsymbol{E}, \boldsymbol{W})$;
Let $\boldsymbol{C}$ be a set of detected communities;
Set $\boldsymbol{C} \leftarrow \emptyset$;
while there are nodes that can be expanded **do**
  Calculate the core dominance measure for each node $\boldsymbol{u_i} \in \boldsymbol{U}, \boldsymbol{i} = \boldsymbol{1}, \boldsymbol{2}, \ldots, |\boldsymbol{U}|$ using Eq. (14);
  Consider a node $\boldsymbol{c} \in \boldsymbol{U}$ with the highest core dominance value as the center of community;
  Set $\boldsymbol{U} \leftarrow \boldsymbol{U} - \{\boldsymbol{c}\}$;
  Set $\boldsymbol{T} \leftarrow \emptyset$;
  for each node $\boldsymbol{u_i} \in \boldsymbol{U}, \boldsymbol{i} = \boldsymbol{1}, \boldsymbol{2}, \ldots, |\boldsymbol{U}|$ **do**
    Calculate the membership degree of node $\boldsymbol{u_i}$ using Eq. (15);
    if (the membership degree of node $\boldsymbol{u_i}$ is greater than 0.5) **then**
      Add node $\boldsymbol{u_i}$ to the community with center $\boldsymbol{c}$;
      Add all the intermediate neighbors of $\boldsymbol{u_i}$ to the community;
      Add $\boldsymbol{u_i}$ and its intermediate neighbors to $\boldsymbol{T}$;
    end if
  end for
  Add the community with center $\boldsymbol{c}$ to the set of detected communities $\boldsymbol{C}$;
  Set $\boldsymbol{U} \leftarrow \boldsymbol{U} - \boldsymbol{T}$;
end while
Add all outlier nodes to the outlier nodes set $\boldsymbol{O}$ if they do not belong to any community;
for each node $\boldsymbol{c_i} \in \boldsymbol{C}, \boldsymbol{i} = \boldsymbol{1}, \boldsymbol{2}, \ldots, |\boldsymbol{C}|$ **do**
  Calculate the embeddedness value of all nodes that belong to the community $\boldsymbol{c_i}$ using Eq. (16);
  Consider a node with the highest embeddedness value as the representative node of the community $\boldsymbol{c_i}$;
end for
for each node $\boldsymbol{o_i} \in \boldsymbol{O}, \boldsymbol{i} = \boldsymbol{1}, \boldsymbol{2}, \ldots, |\boldsymbol{O}|$ **do**
  Merge $\boldsymbol{o_i}$ into the community that the weight between its representative node and $\boldsymbol{o_i}$ is the highest value;
end for
Return the set of detected communities $\boldsymbol{C}$;

Computational Complexity of Algorithm 1. The proposed community detection algorithm needs to perform four main functions in its procedure to detect user communities. Therefore, the computational complexity of this algorithm is strongly dependent on the complexity order of these four main components, which include the calculation of similarity values between

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users and determining the core dominance measure, membership degree, and embeddedness value for all users. These similarity values are obtained using Eq. (12) with a complexity of $O(|U|^2)$. Moreover, the core dominance measure, membership degree, and embeddedness value for each user are determined using Eqs. (14), (15), and (16), respectively. According to these equations, as these metrics are calculated for all users, the complexities of core dominance, membership degree, and embeddedness calculations are $O(|U|^2)$, $O(|U|^2.|C|)$, and $O(|C|.|U|)$, respectively. Notably, $|U|$ and $|C|$ denote the number of users and communities, respectively. The overall computational complexity of Algorithm 1 can be obtained as $O(|U|^2 + |U|^2 + |U|^2.|C| + |C|.|U|)$ by summing the complexities of all four main components. As $|C| \ll |U|$, the overall computational complexity can be reduced to $O(|U|^2)$.

3.4 Virtual rating prediction

This step aims to devise an approach to predict virtual ratings for expanding users' rating profiles. Here, the communities of users detected by the proposed community detection method are utilized to predict virtual ratings. Specifically, users within each community are regarded as nearest neighbors, and their opinions regarding unknown ratings are leveraged in the rating prediction process. Notably, virtual ratings are only predicted for foods that have not yet been rated by the target user. Put differently, foods already rated by the target user are excluded from the virtual rating prediction procedure. Additionally, virtual ratings are predicted solely for foods that have been rated by users belonging to the community of the target user. This helps reduce the computational complexity of the proposed method, as the number of foods rated by users within a community is typically less than the total number of available foods in the system. Consider $C_{u,i}$ as a set of users within the community to which the target user u belongs, who have rated food i. Hence, a virtual rating for food f_i can be predicted using the opinions of these users by the following equation:

$$\mathrm{VR}_{u,i} = \overline{r}_u + \frac{\sum_{v \in C_{u,i}} w_{u,v} \times (r_{v,i} - \overline{r}_v)}{\sum_{v \in C_{u,i}} w_{u,v}} \tag{17}$$

where the average of ratings for user u is represented by $\overline{r}_u$, the rating given by user v to food f_i is denoted by $r_{v,i}$, and Eq. (12) is used to calculate $w_{u,v}$.

3.5 Reliability evaluation

A food recommendation system is a knowledge filtering method to assist users in selecting relevant food menus depending on their dietary habits, diet, and overall health. Thus, examining the reliability and performance of recommendations that consider adequate health indicators is a challenging problem in food recommendation. Furthermore, in order to enhance the rating profiles of users, virtual ratings are employed. Therefore, the healthiness and reliability of the virtual ratings should be taken into account before adding them to the rating profiles, as the food recommendations directly impact the health of users. Thus, we propose a new health-aware reliability measure that can concurrently take health and accuracy aspects into account to evaluate the predicted virtual ratings obtained from Eq. (17). Notably, both health and accuracy aspects are important in the proposed reliability measure, as the food

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Table 2 Nutrition components and their ideal ranges suggested by WHO

Dietary factor	Ideal range
Proteins	10–15%
Carbohydrates	55–75%
Sugars	< 10%
Sodium	< 5 g
Fat	15–30%
Saturated fats	< 10%
Fibers	> 10 g

https://www.who.int/health-topics/nutrition

recommendations should not only be healthy but also accurate to maintain acceptable user satisfaction. Therefore, the health-aware reliability criterion takes into account three factors: The first factor is related to the healthiness of the virtual ratings, whereas the second and third ones are used to address the accuracy of the virtual ratings. The following discussion covers the specifics and ideas behind these three factors.

First factor Seven nutritional components in each food: proteins, carbohydrates, sugars, sodium, fat, saturated fats, and fibers are considered in this factor to ensure the healthiness of the virtual ratings. These nutritional components are selected based on the World Health Organization (WHO) recommendations, which provide the constitutions of these components to ensure healthy and nutritious food. Table 2 shows the recommended healthy ranges of nutrition components by WHO. The main idea of the first factor is that having neighbors who are passionate about healthy eating can improve the accuracy of predicted virtual ratings from a health perspective. To calculate this factor, we first determine the level of healthiness for each food rated by a neighbor $v \in C_{u,i}$. Thereafter, we utilize the summation of the levels of healthiness obtained for all foods rated to determine the healthiness score of v as follows:

$$\mathrm{HU}(v) = \frac{\sum_{i \in I_v} \mathrm{HF}(i)}{|I_v|} \tag{18}$$

where the health level of $v \in C_{u,i}$ is denoted by $\mathrm{HU}(v)$, I_v represents the foods rated by user v, and $\mathrm{HF}(i)$ is measured using the following equation:

$$\begin{aligned} \mathrm{HF}(i) = {} & \mathrm{LabelWHO(proteins)} + \mathrm{LabelWHO(carbohydrates)} \\ & + \mathrm{LabelWHO(sugars)} + \mathrm{LabelWHO(sodium)} + \mathrm{LabelWHO(fat)} \\ & + \mathrm{LabelWHO(saturated\ fats)} + \mathrm{LabelWHO(fibers)} \end{aligned} \tag{19}$$

where, regarding each nutrition element, $\mathrm{LabelWHO}(\dots) = 1$ if its quantity is within the standard range indicated in Table 2; otherwise, $\mathrm{LabelWHO}(\dots) = 0$. Similarly, food is completely unhealthy if its nutrition elements meet none of the standard ranges. In contrast, food is completely healthy if all its nutrition elements are in the ideal ranges reported in Table 2.

When calculating the reliability value of a predicted virtual rating, we consider the health scores of all neighbors of the target user. The first factor related to the reliability value of the

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predicted virtual rating ($VR_{u,i}$) is computed as follows:

$$f_H\left(H_{u,i}\right) = 1 - \frac{\overline{H}}{\overline{H} + H_{u,i}} \tag{20}$$

where $H_{u,i}$ is calculated by summing the health scores of all neighbors of the target user u who have assigned a rating to food f_i and the median of the values of $H_{u,i}$ is denoted by $\overline{H}$. The following formula computes $H_{u,i}$ based on the aforementioned definition:

$$H_{u,i} = \sum_{v \in C_{u,i}} \mathrm{HU}(v) \tag{21}$$

where $C_{u,i}$ represents a subset of users whose community is the same as that of the target user u and have assigned a rating to food f_i, and $\mathrm{HU}(v)$ denotes the healthiness score of user v obtained from Eq. (18). Equation (20) reveals that the first factor impacts the reliability measure positively, and any increment in this factor increases the reliability from a health perspective, and vice versa.

Second factor In addition to the health criterion, the food recommendations should be relevant to the preferences of users, as making irrelevant (i.e., inaccurate) recommendations lowers the efficiency of RS by deteriorating the satisfaction level of users. As a result, we should consider the accuracy-related factors in the proposed reliability measure to enhance its quality in evaluating the virtual ratings. According to Eq. (17), the similarity values between users are vital in predicting virtual ratings. Thus, these values can be utilized as a metric that positively affects the reliability measure, meaning that if the similarity value between two users is high, then their opinions are similar. In other words, predicting virtual ratings using more similar neighbors leads to a more accurate prediction and increases the reliability from an accuracy perspective. The second factor constitutes a component of the proposed reliability measure, calculated by summing the similarity values between the target user and all neighbors taken into account during the virtual rating prediction process. This factor is calculated using the following formula:

$$f_S\left(S_{u,i}\right) = 1 - \frac{\overline{S}}{\overline{S} + S_{u,i}} \tag{22}$$

where $S_{u,i}$ represents the summation of similarity values between the target user u and all neighbors employed in predicting the virtual rating $VR_{u,i}$, and $\overline{S}$ is the median of $S_{u,i}$ values. Equation (23) is used to obtain $S_{u,i}$ as follows:

$$S_{u,i} = \sum_{v \in C_{u,i}} w_{u,v} \tag{23}$$

Third factor This factor is used to measure the reliability value of a virtual rating based on the variance of the ratings utilized in predicting the virtual rating. How similar the opinions of neighbors are is crucial. If the neighbors have very different opinions about an item (i.e., a high level of variance), a virtual rating with low reliability can be expected. On the other hand, predicting a virtual rating based on a set of similar opinions (i.e., a low level of variance) will result in higher reliability. Therefore, we can conclude that the third factor negatively

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impacts the reliability level of a virtual rating. Equation (24) defines the third factor:

$$f_V\left(V_{u,i}\right) = \left(\frac{\text{maxR} - \text{minR} - V_{u,i}}{\text{maxR} - \text{minR}}\right)^{\gamma} \tag{24}$$

where we represent the minimum and maximum values of ratings assignable to foods as minR and maxR, respectively, and $V_{u,i}$ is defined as:

$$V_{u,i} = \frac{\sum_{v \in C_{u,i}} w_{u,v} \times (r_{v,i} - \overline{r}_v - VR_{u,i} + \overline{r}_u)^2}{\sum_{v \in C_{u,i}} w_{u,v}} \tag{25}$$

and γ is defined as:

$$\gamma = \frac{\ln 0.5}{\ln \frac{\text{maxR} - \text{minR} - \overline{V}}{\text{maxR} - \text{minR}}} \tag{26}$$

where $\overline{V}$ indicates the median of $V_{u,i}$ values.

Now, we define the proposed health-aware reliability measure by taking the three above-mentioned factors into account and using the geometric average function to integrate them into a unique measure. For this purpose, we need to establish a weight for each factor indicating the dependencies between these three factors in the geometric average function. Noticeably, the first factor $f_H\left(H_{u,i}\right)$ and the second factor $f_S\left(S_{u,i}\right)$ have no dependencies on any other factors. This means that these factors are completely independent, and their weights should be set to 1. However, the third factor $f_V\left(V_{u,i}\right)$ is not independent, and its value can be affected by the first factor $f_S\left(S_{u,i}\right)$. As a result, we set the weight of $f_V\left(V_{u,i}\right)$ in the geometric average function to $f_S\left(S_{u,i}\right)$. By integrating the defined factors and considering their determined weights in the geometric average function, the reliability measure is obtained for a virtual rating $VR_{u,i}$:

$$\text{Rel}_{u,i} = \left(f_H\left(H_{u,i}\right).f_S\left(S_{u,i}\right).f_V\left(V_{u,i}\right)^{f_S\left(S_{u,i}\right)}\right)^{\frac{1}{2+f_S\left(S_{u,i}\right)}} \tag{27}$$

Equation (27) not only considers the health factor but also two accuracy-related factors when evaluating predicted virtual ratings. Therefore, the proposed measure can be effective in enhancing food RSs by ensuring they make relevant and healthy food recommendations for users.

3.6 Rating profile expansion

This step expands the rating profiles of users using a mechanism that adds some virtual ratings to profiles with insufficient ratings to make accurate predictions. The proposed probabilistic model described in Sect. 3.2 plays a vital role in expanding the rating profiles effectively, as it helps to identify the profiles that need to be expanded. Notably, expanding the profiles whose own real ratings are sufficient to make accurate recommendations may lower their effectiveness significantly. Therefore, we first compare the number of real ratings of the target user (i.e., $|I_u|$) to the minimum number of required ratings (i.e., n'_u). Then, we only expand the rating profile if the number of real ratings is less than the minimum value (i.e., $|I_u| < n'_u$). Two important issues are considered in the proposed rating profile expansion technique. First, the virtual ratings are calculated using Eq. (17) only for the foods that are among the preferences of the users within the target user's community. This lowers the

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computational complexity of the proposed method, as the number of foods preferred by the neighbors is much lower than the whole number of foods. Additionally, those foods that have been already rated by the target user are not considered in the virtual rating prediction. In other words, the real ratings are employed without any changes along with the added virtual ratings in the final rating prediction process. Second, the reliability of the virtual ratings calculated using Eq. (27) is used as a metric to select the virtual ratings to be added to the target user's profile. As a result, we sort all predicted virtual ratings based on their reliability values in descending order and select $n'_u - |I_u|$ ratings for expanding the user u' rating profile. This can guarantee that the added virtual ratings are healthy and accurate, as both these criteria are covered in the proposed reliability measure. As the expanded rating profiles contain more ratings than their original ones, the user–food interaction matrix will also be denser than the original one. Therefore, the proposed rating profile expansion approach not only considerably alleviates the data sparsity and cold start problems but also helps food RSs to make more healthy and accurate recommendations to users.

3.7 Recommendation

In this step, instead of using the users' original rating profiles, we use their expanded ones to make food recommendations. In other words, the expanded user–food interaction matrix that is denser than the original is employed in the calculation of similarity values. To do this, we first use the time-aware similarity function represented in Eq. (12) to compute the similarity values. Notably, unlike the real ratings, the virtual ratings added to the rating profile of the target user do not have any timestamps. However, according to Eq. (12), we need their timestamps to compute the similarity values. The virtual ratings timestamps are calculated by taking the mean of all the ratings' timestamps expressed by the target user. By computing similarity values, a nearest neighbor set is formed for the target user by selecting N users with highest similarity values. This set is utilized as follows to predict the unknown rating for the user u and the food f_i:

$$P_{u,i} = \overline{r}_u + \frac{\sum_{v \in NN_{u,i}} w_{u,v}\left(r_{v,i} - \overline{r}_v\right)}{\sum_{v \in NN_{u,i}} w_{u,v}} \tag{28}$$

where $NN_{u,i}$ refers to the nearest neighbors of the target user u, $\overline{r}_u$ is the average of all ratings expressed by user u, $r_{v,i}$ is used to denote the rating of food f_i assigned by user v, and $w_{u,v}$ is calculated using Eq. (12) as a similarity score between users u and v. As the output, the proposed method provides a list of food recommendations to users by choosing a subset of foods whose predicted ratings are higher. The primary difference between the proposed recommendation process and other previous food recommendation models is that we employ a user–food interaction matrix expanded by virtual ratings instead of the original one. Moreover, the virtual ratings are selected based on a new health-aware reliability measure, which simultaneously considers the health and accuracy factors. Therefore, the proposed method helps the users receive healthier and relevant food recommendations. Moreover, the proposed method helps the users toward a healthier diet. The overall procedure of the HRRPE model is summarized in Algorithm 2.

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Algorithm 2 Healthy and Reliable Rating Profile Expansion Approach for Food RSs (HRRPE)

Inputs: User-food matrix ($\boldsymbol{R}$), recommendations list length ($\boldsymbol{L}$), $\boldsymbol{\delta}$, $\boldsymbol{\lambda}$, and number of nearest neighbors ($\boldsymbol{N}$).
Output: Food recommendations.

Split dataset into two subsets $\boldsymbol{Tr} \subset \boldsymbol{R}$ and $\boldsymbol{Te} \subset \boldsymbol{R}$ as the training set and test set, correspondingly;
Suppose $\boldsymbol{U}$ and $\boldsymbol{I}$ are the users set and foods set, respectively;
for each user $\boldsymbol{u} \in \boldsymbol{U}$ **do**
    Obtain the value of $\boldsymbol{n'_u}$ using Eq. (3);
end for
Obtain the user communities using Algorithm 1 based on $\boldsymbol{Tr}$;
for each $\boldsymbol{u} \in \boldsymbol{U}$ **do**
    Suppose $\boldsymbol{I_u}$ denotes a set of foods rated by user $\boldsymbol{u}$;
    if $|\boldsymbol{I_u}| < \boldsymbol{n'_u}$ **then**
        Suppose $\boldsymbol{C_u}$ is the community of user $\boldsymbol{u}$;
        Suppose $\boldsymbol{I_{C_u}}$ is a set of foods rated by the users of $\boldsymbol{C_u}$;
        Let $\overline{\boldsymbol{I}}_{\boldsymbol{u}} = \boldsymbol{I_{C_u}} - \boldsymbol{I_u}$;
        for each food $\boldsymbol{i} \in \overline{\boldsymbol{I}}_{\boldsymbol{u}}$ **do**
            Predict the virtual rating $\boldsymbol{VR_{u,i}}$ using Eq. (17);
            Compute the reliability value $\boldsymbol{Rel_{u,i}}$ for the virtual rating $\boldsymbol{VR_{u,i}}$ using Eq. (27);
        end for
        Expand the rating profile of the user $\boldsymbol{u}$ by adding $\boldsymbol{n'_u} - |\boldsymbol{I_u}|$ virtual ratings with the highest reliability values;
    end if
end for
for each $\boldsymbol{u} \in \boldsymbol{U}$ **do**
    Calculate time-aware similarity values using Eq. (12) based on the expanded rating profiles of users;
    Form the nearest neighbors set by selecting $\boldsymbol{N}$ users with the highest similarity values;
    for each $\boldsymbol{r_{u,i}} \in \boldsymbol{Te}$ where $\boldsymbol{i} \in \boldsymbol{I}$ **do**
        Predict the unknown rating $\boldsymbol{P_{u,i}}$ using Eq. (28);
    end for
    Provide a recommendations list to user $\boldsymbol{u}$ by selecting $\boldsymbol{L}$ foods with the highest ratings;
end for

3.8 Computational complexity

Besides evaluating the performance of a recommendation method in generating accurate recommendations, analyzing computational complexity is crucial for assessing the method's feasibility in real-world applications. This section aims to evaluate and analyze the computational complexity of the HRRPE model. Thus, we begin by assessing the complexity of each stage of HRRPE and subsequently compute the overall complexity by aggregating the complexities of all stages. The initial step, rating profile evaluation, can be executed with a complexity of $O(|U||I|)$, where $|U|$ and $|I|$ denote the number of users and foods, respectively. In Sect. 3.3, we demonstrated that the complexity of the second step (the proposed community detection algorithm) is $O\left(|U|^2\right)$. The third step of HRRPE is performed with a complexity of $O(|U||I|)$. The fourth step of HRRPE is related to the reliability evaluation of the virtual ratings, which can be performed with the complexity of $O\left(|U|^2|I|\right)$. In the fifth step, the expansion procedure of the rating profiles is conducted, necessitating the sorting of virtual ratings. Thus, the fifth step's complexity is $O(|U||I|\log|I|)$. In the ultimate step of HRRPE, the recommendation procedure is employed to generate food recommendations, with a requisite complexity of $O(|U||I|)$. After determining the complexities of all steps of HRRPE, its overall complexity is obtained as $O\left(3|U||I| + |U|^2 + |U|^2|I| + |U||I|\log|I|\right)$. Thus, the reduced complexity is obtained as $O\left(|U|^2|I|\right)$.

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4 Experiments

In this section, we evaluate our HRRPE model by conducting extensive experiments on two food datasets to verify its effectiveness compared to other food RSs. Below, we provide a brief overview of the recommendation approaches utilized in the experiments for comparison with our developed HRRPE:

- *LDA* [46]: A food RS that evaluates the nutritional quality of recipes and suggests healthier options.
- *HPRR* [57]: A food RS that generates both healthy and accurate recommendations by incorporating a health bias and a health tag in the food RS.
- *CNNFR* [58]: A system utilizing deep convolutional neural networks to recognize the components of recipes and utilize these components for crafting an effective food recommendation system.
- *HAFR* [43]: A visual-aware food RS based on a hierarchical attention network and deep learning, which utilizes the user's historical ratings, food content, and food images.
- *CFRR* [59]: An integrated health-aware food RS that combines collaborative filtering and content-based models. To generate more healthy recommendations, it filters out foods that contain more calories than the required daily calorie intake for the intended user.
- *FGCN* [44]: A deep learning-based food RS based on ingredients–ingredient, ingredients–recipe, and recipes–user relations in three networks generated by a graph convolutional network.
- *HGAT* [60]: A food RS based on a hierarchical graph-based attention network optimized by a ranking-based objective function to make recommendations according to the users' preferences and foods' ingredients.

4.1 Datasets

Two food RS datasets, www.allrecipes.com and www.food.com, were utilized to evaluate the compared recommendation models. In the Allrecipes.com dataset [60], 52,821 recipes were extracted from 27 different food categories for the period 2000–2018. For each food, users' ratings, nutrition information, and rating timestamps were crawled. The information related to 68,768 users, 45,630 foods, and 1,093,845 ratings was obtained after preprocessing this dataset. The Food.com dataset [26] contains 25,076 users, 178,265 foods, and 749,053 ratings collected from 2000 to 2018. To analyze the healthiness of foods, seven nutrition components—proteins, carbohydrates, sugars, sodium, fat, saturated fats, and fibers—that are available in the Allrecipes.com dataset were utilized for each food. In the Food.com dataset, the information related to the nutrition component fibers was not available; thus, we supposed that the value of this component for all foods in this dataset is in the ideal range. In both datasets, the ratings assigned to foods by users were in the range of 1–5. The detailed descriptions of these datasets are listed in Table 3.

4.2 Evaluation metrics

In this study, four evaluation criteria—precision, recall, normalized discounted cumulative gain (NDCG), and health score—were utilized to investigate the RS performance of the compared systems. Precision, recall, and NDCG are popular metrics that are extensively utilized to assess the effectiveness of RSs based on the accuracy of the recommendation

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Table 3 Statistics of Allrecipes.com and Food.com datasets

Characteristics	Allrecipes.com	Food.com
#Users	68,768	25,076
#Foods	45,630	178,265
#Ratings	1,093,845	749,053
Average rating/per user	15.90	29.87
Average rating/per item	23.97	4.2
Sparsity (%)	99.96	99.98

list generated by these systems. Precision and recall metrics assess RSs by quantifying the proportion of relevant recommendations provided to users as follows:

$$\text{precision} = \frac{|\{\text{recommended items that are relevant}\}|}{L} \tag{29}$$

$$\text{recall} = \frac{|\{\text{recommended items that are relevant}\}|}{|\{\text{all relevant items}\}|} \tag{30}$$

where L determines the length of the recommendations list (i.e., the number of foods recommended to users). The ability of recommendation models to generate a list of recommendations featuring relevant items at the top of the list is investigated using NDCG metric as follows:

$$\text{NDCG} = \frac{\text{DCG}}{\text{DCG}_{\text{max}}} \tag{31}$$

where

$$\text{DCG} = \text{rel}_1 + \sum_{i=2}^{L} \frac{\text{rel}_i}{\log_2(i+1)} \tag{32}$$

and rel_i is employed to specify whether the recommended food f_i is relevant. Thus, if the suggested item f_i is relevant, then $\text{rel}_i = 1$; otherwise, $\text{rel}_i = 0$. Moreover, DCG_{max} is defined as below:

$$\text{DCG}_{\text{max}} = 1 + \sum_{i=2}^{L} \frac{1}{\log_2(i+1)} \tag{33}$$

In addition to the aforementioned accuracy criteria, we defined a novel measure for evaluating the recommendation list based on the recommended foods' health levels. Based on this metric, how healthy the recommended foods are can be determined as follows:

$$\text{Health_score} = \frac{\sum_{i=1}^{L} \text{HF}(f_i)}{7 \times L} \tag{34}$$

where $\text{HF}(f_i)$ indicates the health aspect of item f_i, which can be computed using Eq. (19). The health-score criterion's value can range from 0 to 1, wherein a Health_score $= 0$ means that the suggested recipes are completely unhealthy, whereas Health_score $= 1$ denotes that the suggested foods are completely healthy.

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4.3 Parameter setting and analysis

In evaluating an algorithm, tuning the input parameters to reach the highest efficiency is important. In this section, our objective was to fine-tune the input parameters of the proposed HRRPE method by performing a sensitivity analysis, aiming to ascertain the optimal values for these parameters. The proposed HRRPE method comprises three input parameters: (1) δ is utilized in Eq. (3) as a predefined threshold to determine the uncertainty level of the predictions, (2) λ is a parameter used in Eq. (13) to control the effect of the time factor in calculating the time-aware similarity values, and (3) N is the third input parameter referring to the number of nearest neighbors employed in the recommendation process. As part of our experiments, we explored the impact of varying values of these parameters on the performance of HRRPE. Figure 2 shows the sensitivity analysis of parameter δ under different evaluation metrics and $L = 10$ where the value of this parameter is changed from 0.1 to 0.9 with a step size of 0.1. Noticeably, the performance of HRRPE declines in most cases when the value of δ is increased. This is expected because higher values of δ create a higher level of uncertainty in the rating prediction process. On the other hand, when δ is less than a threshold value, the performance of HRRPE is reduced significantly. In other words, the performance of HRRPE declined when δ was less than 0.3 and 0.2 for the Allrecipes.com and Food.com datasets, respectively. This is due to adding some irrelevant virtual ratings to the rating profiles of users in the rating profile expansion step. As a result, we chose $\delta = 0.3$ for the Allrecipes.com dataset and $\delta = 0.2$ for the Food.com dataset as the best values to compare the proposed HRRPE method to other models.

The sensitivity analysis of parameter λ is illustrated in Fig. 3, where we set its value from $\lambda = 0$ to $\lambda = 4$ with a step size of 0.5. Notably, $\lambda = 0$ means that we ignore the time factor when calculating the similarity values between users. Figure 3 demonstrates

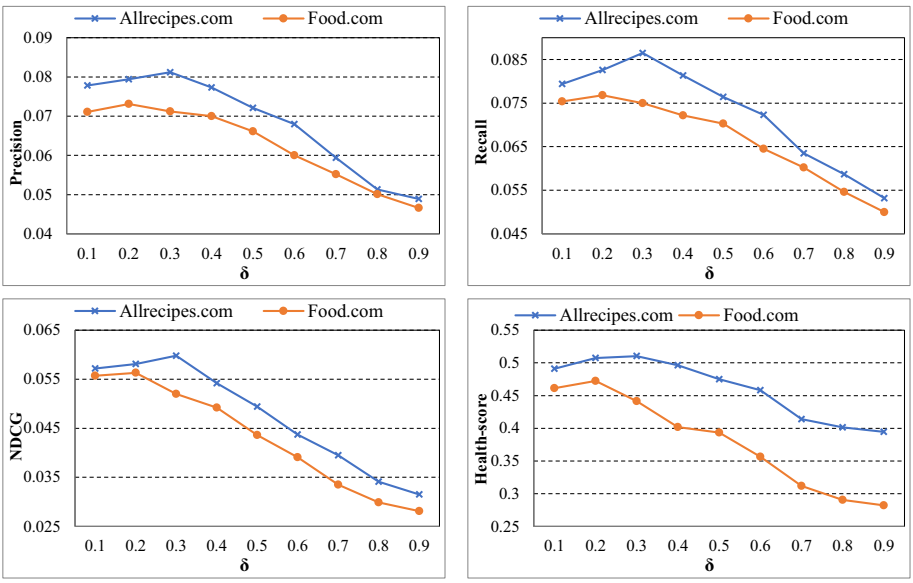


Fig. 2 Evaluation of the performance of the proposed HRRPE method with different values of parameter δ and length of recommendations list $L = 10$ for Allrecipes.com and Food.com datasets

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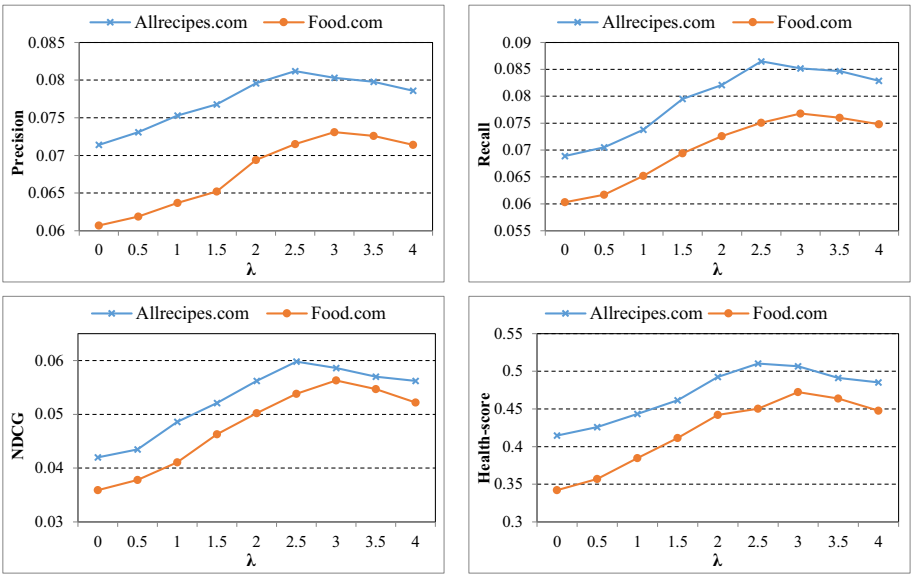


Fig. 3 Evaluation of the performance of the proposed HRRPE method with different values of parameter λ and the length of recommendations list $L = 10$ for Allrecipes.com and Food.com datasets

that the time factor has a positive effect on the performance of HRRPE, as the values of all evaluation metrics increase when the value of λ increases. Therefore, considering the changes in user's diet over time in the recommendation process increases the accuracy of food recommendations. However, these results demonstrate that the performance diminishes when λ exceeds 2.5 and 3 for the Allrecipes.com and Food.com datasets, respectively. This is attributed to the recommendations being only provided based on the recent preferences of users when λ exceeds these values, and most of their historical data are ignored by the RS. As a result, ensuring a balance for this parameter is necessary to maintain the proposed HRRPE method at the highest level of efficiency. In the experiments, we initialized $\lambda = 2.5$ for the Allrecipes.com dataset and $\lambda = 3$ for the Food.com dataset to evaluate the performance of the proposed HRRPE method compared to other recommender systems. The third aspect of the sensitivity analysis concerns parameter N, and the experimental findings of this analysis are depicted in Fig. 4. To carry out these experiments, we examined various values of N (i.e., different numbers of nearest neighbors), namely $N = 20$, $N = 50$, $N = 70$, $N = 90$, $N = 120$, and $N = 150$. Notably, adjusting the value of N has a considerable effect on the quality of recommendations. In other words, considering a small number of nearest neighbors in the recommendation process leads to inaccurate predictions. Similarly, an excessively large number of nearest neighbors decreases the quality of recommendations due to irrelevant opinions being incorporated into the recommendation process. The results shown in Fig. 4 indicate that the best values are $N = 70$ for the Allrecipes.com dataset and $N = 90$ for the Food.com dataset, as the proposed HRRPE method achieves the best performance in these cases. Therefore, we chose these values for parameter N in the experiments to compare the effectiveness of the proposed HRRPE method to other recommendation models. Notably, all experiments were performed by splitting the datasets into two sets, namely training and test sets, where the earliest 80% of ratings according to the available timestamps were considered as the training set, and the remaining 20% were utilized as the test set.

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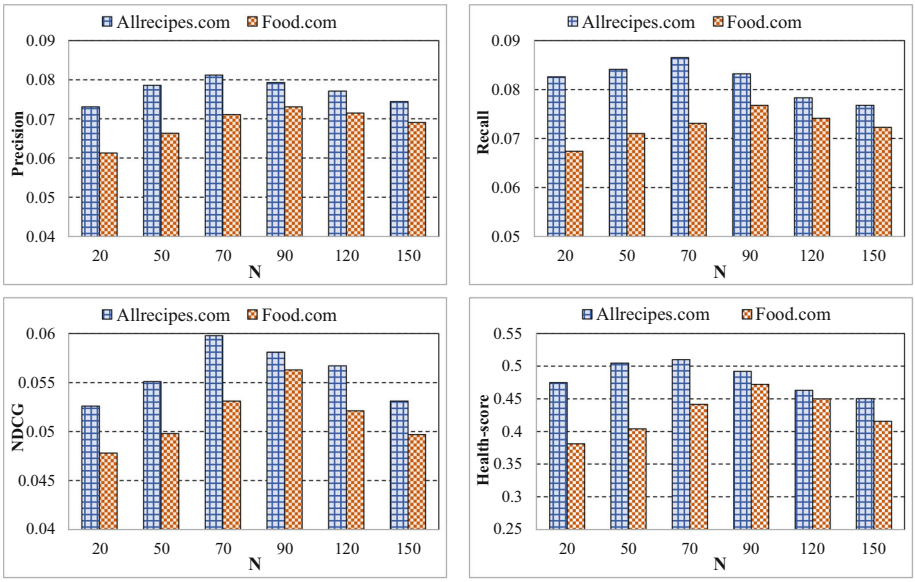


Fig. 4 An evaluation of the performance of the developed HRRPE method with different values of parameter N and the length of recommendations list $L = 10$

4.4 Performance comparison

The proposed HRRPE method was evaluated in this section and compared to other recommendation approaches by utilizing various evaluation metrics. Additionally, we considered different recommendation list lengths (L), as the calculation of the evaluation metrics is dependent on the value of L. In this regard, the precision, recall, and NDCG metrics were employed to assess the accuracy of the food recommendations generated by the compared models. In contrast, the health-score metric evaluates the generated food recommendations list according to the health level of foods. The experimental results related to the precision, recall, and NDCG metrics for the Allrecipes.com and Food.com datasets are tabulated in Tables 4 and 5, respectively. These findings demonstrate that the proposed HRRPE method surpassed other models, achieving the highest precision, recall, and NDCG metrics. As a part of the results reported in Table 4, for $L = 10$, HRRPE achieves values of 0.0812, 0.0865, and 0.0598 for the precision, recall, and NDCG metrics, respectively, whereas for the same case, HGAT achieved values of 0.0721, 0.0746, and 0.0497 as the second-best performer. As a result, the improvements made by the proposed method compared to the second-best model were approximately 12.62%, 15.95%, and 20.32% for the precision, recall, and NDCG metrics, respectively. Moreover, Table 5 demonstrates that HRRPE is better than the second-best model by approximately 9.92%, 12.11%, and 17.78% for $L = 10$ based on the precision, recall, and NDCG metrics, respectively, for the Food.com dataset. Hence, Tables 4 and 5 highlight that HRRPE can notably outperform other recommendation models across all employed metrics and datasets.

Another comparison was conducted based on the health-score metric, and the results are reported in Table 6 for the Allrecipes.com and Food.com datasets. According to the results reported in Table 6, HRRPE is the best model and achieves health levels of 0.5103,

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Table 4 Evaluation of the proposed HRRPE method compared to other RSs on the Allrecipes.com dataset across precision, recall, and NDCG metrics, considering varying lengths of recommendation list (L)

Methods	Precision			Recall			NDCG		
	$L = 10$	$L = 15$	$L = 20$	$L = 10$	$L = 15$	$L = 20$	$L = 10$	$L = 15$	$L = 20$
LDA	0.0582	0.0451	0.0326	0.0604	0.0697	0.0789	0.0381	0.0439	0.0463
HPRR	0.0596	0.0478	0.0367	0.0635	0.0732	0.0816	0.0402	0.0428	0.0476
CNNFR	0.0625	0.0509	0.0398	0.0652	0.0713	0.0864	0.0411	0.0446	0.0492
HAFR	0.0698	0.0512	0.0409	0.0672	0.0859	0.1021	0.0452	0.0491	0.0512
CFRR	0.0674	0.0571	0.0492	0.0648	0.0881	0.1064	0.0437	0.0551	0.0591
FGCN	0.0706	0.0582	0.0538	0.0682	0.0894	0.1102	0.0466	0.0609	0.0716
HGAT	0.0721	0.0642	0.0557	0.0746	0.0928	0.1176	0.0497	0.0621	0.0743
HRRPE	0.0812	0.0794	0.0723	0.0865	0.1051	0.1293	0.0598	0.0704	0.0796

Table 5 Evaluation of the proposed HRRPE method compared to other RSs on the Food.com dataset across precision, recall, and NDCG metrics, considering varying lengths of recommendation list (L)

Methods	Precision			Recall			NDCG		
	$L = 10$	$L = 15$	$L = 20$	$L = 10$	$L = 15$	$L = 20$	$L = 10$	$L = 15$	$L = 20$
LDA	0.0491	0.0372	0.0264	0.0514	0.0589	0.0692	0.0314	0.0359	0.0382
HPRR	0.0503	0.0405	0.0289	0.0528	0.0634	0.0764	0.0345	0.0367	0.0426
CNNFR	0.0542	0.0446	0.0321	0.0562	0.0682	0.0796	0.0367	0.0392	0.0411
HAFR	0.0608	0.0471	0.0335	0.0633	0.0793	0.0879	0.0432	0.0489	0.0508
CFRR	0.0587	0.0498	0.0397	0.0612	0.0746	0.0913	0.0391	0.0461	0.0531
FGCN	0.0634	0.0512	0.0443	0.0654	0.0811	0.1024	0.0446	0.0546	0.0629
HGAT	0.0665	0.0576	0.0472	0.0685	0.0864	0.1078	0.0478	0.0572	0.0682
HRRPE	0.0731	0.0689	0.0598	0.0768	0.0972	0.1135	0.0563	0.0665	0.0729

Table 6 Evaluation of the proposed HRRPE method compared to other RSs in terms of health-score metric and different lengths of recommendations list (L)

Methods	Allrecipes.com			Food.com		
	$L = 10$	$L = 15$	$L = 20$	$L = 10$	$L = 15$	$L = 20$
LDA	0.3827	0.3601	0.3368	0.4016	0.3714	0.3334
HPRR	0.4291	0.4113	0.3846	0.4197	0.3998	0.3482
CNNFR	0.2541	0.2312	0.1983	0.2185	0.1937	0.1646
HAFR	0.2713	0.2634	0.2472	0.2437	0.2289	0.2178
CFRR	0.3483	0.3271	0.2924	0.3531	0.3341	0.2923
FGCN	0.3148	0.2914	0.2686	0.3256	0.2859	0.2612
HGAT	0.3017	0.2765	0.2497	0.2993	0.2711	0.2304
HRRPE	0.5103	0.4812	0.4534	0.4724	0.4423	0.3967

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0.4812, and 0.4534 for the Allrecipes.com dataset for $L = 10$, $L = 15$, and $L = 20$, respectively. HPRR achieves the second-best health levels of 0.4291, 0.4113, and 0.3846 for the Allrecipes.com dataset in terms of $L = 10$, $L = 15$, and $L = 20$, respectively. Therefore, HRRPE outperformed the second-best performer on the Allrecipes.com dataset by approximately 18.92%, 16.99%, 17.88% for $L = 10$, $L = 15$, and $L = 20$, respectively. In addition, Table 6 reports the superiority of the proposed HRRPE method over other models in the case of the Food.com dataset where its performance improvements compared to the second-best model are approximately 12.55%, 10.63%, and 13.92% for $L = 10$, $L = 15$, and $L = 20$, respectively. Therefore, we can conclude that the proposed HRRPE method not only improves the accuracy of food recommendations significantly, but it can also make more healthy recommendations compared to other food recommenders. This achievement demonstrates the importance of the proposed rating profile expansion technique in boosting food recommender systems to make more accurate and healthy recommendations. Further, the proposed health-aware reliability measure plays a vital role in this achievement, as it ensures a balance between the accuracy and health-score metrics to expand the rating profiles.

4.5 Data sparsity and cold start analysis

One of the principal advantages of the proposed HRRPE approach is to address issues of data sparsity and cold start by making more efficient rating profiles for users by expanding them through healthy and reliable virtual ratings. In this section, we design some experimental scenarios to demonstrate how the proposed model improves the recommendation performance when these problems affect the quality of RSs. To do this, we first define a parameter named θ to represent the percentage of data sparsity in the original datasets. Accordingly, five different levels of data sparsity were considered in the experiments. Each level delineates the proportion of the input data utilized as the training set. For instance, $\theta = 10\%$ implies that 90% of the input data is allocated for the training set, while the remaining 10% is designated for the test set. Therefore, a higher value of θ leads to a higher level of data sparsity. We analyzed the performance of different recommendation models in addressing the data sparsity problem by measuring their recommendation quality based on precision, recall, NDCG, and health-score metrics in terms of the defined sparsity levels. The results of this analysis are shown in Figs. 5 and 6 for the Allrecipes.com and Food.com datasets, respectively. Analysis of these findings reveals that HRRPE exhibits superior capability in mitigating the data sparsity issue compared to other models, evidenced by higher values of all evaluation metrics obtained by HRRPE in contrast to others. In other words, HRRPE achieved the highest values of precision, recall, NDCG, and health-score metrics in terms of all data sparsity levels and both datasets. This accomplishment can be attributed to the efficacy of the proposed rating profile expansion technique, which substantially aids the food RS in mitigating the adverse impact of the data sparsity issue. Hence, integrating dependable virtual ratings into users' rating profiles leads to a denser user-food rating matrix compared to the original, thereby facilitating the generation of healthier and more accurate food recommendations through the proposed method. Figures 5 and 6 further illustrate that elevating the data sparsity level (i.e., higher values of θ) diminishes the recommendation performance of all compared models across all evaluation metrics for both datasets. This outcome is anticipated, as accurate recommendations become more challenging with increased data sparsity.

In addition to addressing the data sparsity problem, we also examined the capability of various recommendation models to generate food recommendations for cold start users. In our experiments, we defined cold start users as those whose number of ratings assigned to

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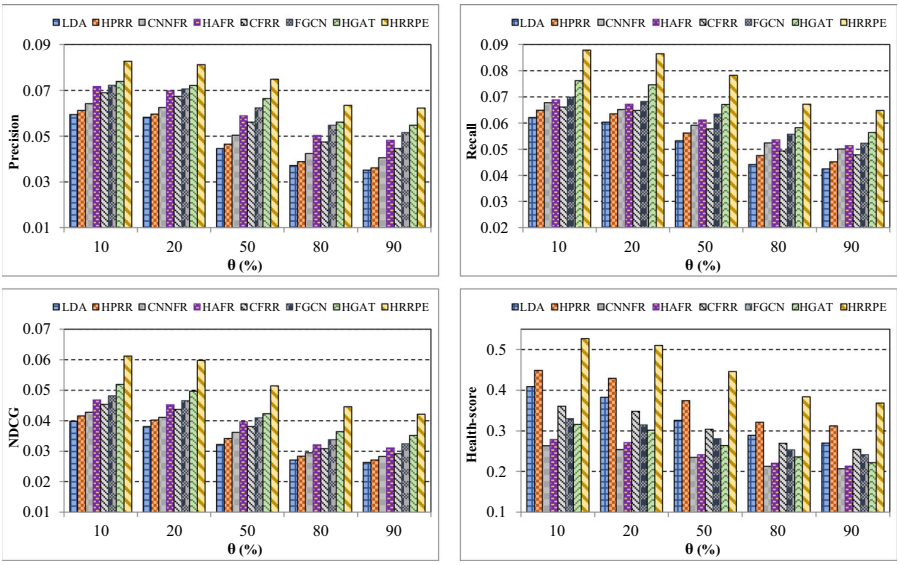


Fig. 5 Comparison of performance in addressing the data sparsity problem based on precision, recall, NDCG, and health-score metrics, different sparsity levels (θ), and the length of recommendations list $L = 10$ for the Allrecipes.com dataset

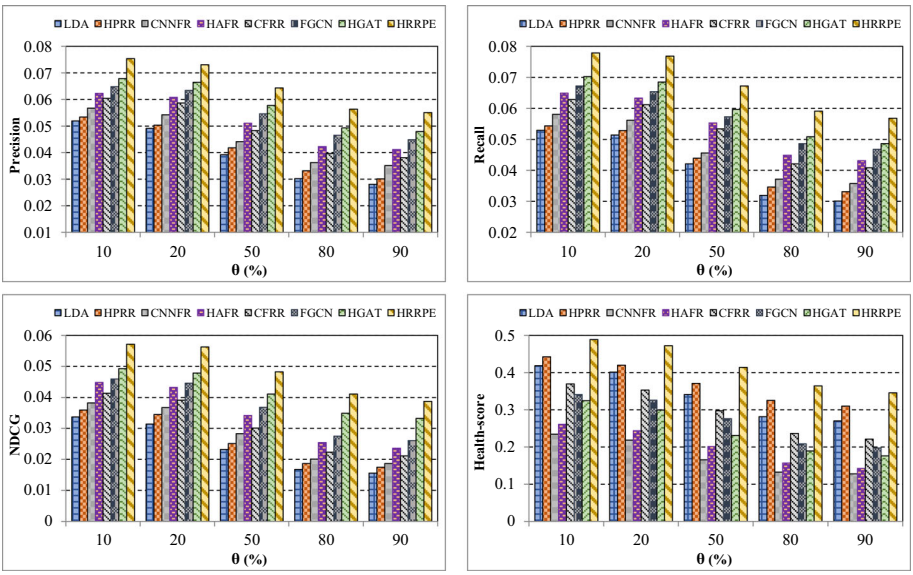


Fig. 6 Comparison of performance in addressing data sparsity problem based on precision, recall, NDCG, and health-score metrics, different sparsity levels (θ), and the length of recommendations list $L = 10$ for the Food.com dataset

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the foods was at most 5. In other words, users who have rated at most 5 foods were considered cold start users. Notably, making accurate recommendations for these users is very challenging for an RS, as there are insufficient ratings to be used in the recommendation process in these cases. Therefore, this analysis is vital for proving how the proposed method can handle this challenge in the context of recommendation systems. Tables 7 and 8 report the results of experiments conducted on the Allrecipes.com and Food.com datasets, respectively, and compare different RSs in addressing cold start based on precision, recall, and NDCG measures. Noticeably, the proposed HRRPE method achieves the best outputs for all lengths of recommendation lists. For instance, Table 7 demonstrates that the precision, recall, and NDCG values for $L = 10$ calculated according to the recommendations generated by HRRPE were 0.0527, 0.0649, and 0.0463, respectively. These values are higher than those of the other recommendation models, where the second-best results are achieved by HGAT: 0.0443, 0.0512, and 0.0358 for precision, recall, and NDCG metrics, respectively. Thus, HRRPE enhances the recommendation performance based on precision, recall, and NDCG metrics by approximately 18.96%, 26.75%, and 29.32%, respectively. Table 8 also

Table 7 Performance comparison of different RSs in addressing the cold start problem for the Allrecipes.com dataset across precision, recall, and NDCG metrics, considering varying lengths of recommendation list (L)

Methods	Precision			Recall			NDCG		
	$L = 10$	$L = 15$	$L = 20$	$L = 10$	$L = 15$	$L = 20$	$L = 10$	$L = 15$	$L = 20$
LDA	0.0231	0.0204	0.0164	0.0326	0.0339	0.0352	0.0165	0.0189	0.0213
HPRR	0.0239	0.0213	0.0191	0.0354	0.0371	0.0405	0.0201	0.0243	0.0264
CNNFR	0.0274	0.0238	0.0211	0.0387	0.0418	0.0436	0.0237	0.0258	0.0273
HAFR	0.0357	0.0264	0.0234	0.0413	0.0422	0.0459	0.0292	0.0317	0.0346
CFRR	0.0339	0.0312	0.0277	0.0372	0.0435	0.0462	0.0274	0.0334	0.0379
FGCN	0.0395	0.0346	0.0309	0.0458	0.0486	0.0527	0.0321	0.0361	0.0404
HGAT	0.0443	0.0392	0.0348	0.0512	0.0547	0.0573	0.0358	0.0399	0.0452
HRRPE	0.0527	0.0476	0.0435	0.0649	0.0694	0.0716	0.0463	0.0485	0.0531

Table 8 Performance comparison of different RSs in addressing the cold start problem for the Food.com dataset across precision, recall, and NDCG metrics, considering varying lengths of recommendation list (L)

Methods	Precision			Recall			NDCG		
	$L = 10$	$L = 15$	$L = 20$	$L = 10$	$L = 15$	$L = 20$	$L = 10$	$L = 15$	$L = 20$
LDA	0.0234	0.0168	0.0136	0.0272	0.0326	0.0368	0.0138	0.0173	0.0198
HPRR	0.0265	0.0201	0.0164	0.0289	0.0374	0.0412	0.0157	0.0182	0.0225
CNNFR	0.0297	0.0237	0.0201	0.0317	0.0413	0.0453	0.0191	0.0246	0.0278
HAFR	0.0351	0.0253	0.0218	0.0415	0.0461	0.0504	0.0269	0.0278	0.0311
CFRR	0.0326	0.0284	0.0263	0.0386	0.0479	0.0547	0.0223	0.0281	0.0339
FGCN	0.0374	0.0327	0.0291	0.0449	0.0521	0.0573	0.0314	0.0345	0.0403
HGAT	0.0411	0.0379	0.0338	0.0473	0.0543	0.0628	0.0352	0.0398	0.0456
HRRPE	0.0513	0.0451	0.0406	0.0559	0.0628	0.0695	0.0413	0.0479	0.0562

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Table 9 Performance comparison of different RSs in addressing cold start problem in terms of health-score metric and different lengths of recommendations list (L)

Methods	Allrecipes.com			Food.com		
	$L = 10$	$L = 15$	$L = 20$	$L = 10$	$L = 15$	$L = 20$
LDA	0.3092	0.2837	0.2592	0.3214	0.3066	0.2703
HPRR	0.3651	0.3362	0.3041	0.3575	0.3306	0.2941
CNNFR	0.1923	0.1685	0.1395	0.1634	0.1422	0.1169
HAFR	0.2164	0.1891	0.1607	0.1876	0.1557	0.1322
CFRR	0.2786	0.2548	0.2269	0.2976	0.2638	0.2388
FGCN	0.2512	0.2318	0.2035	0.2589	0.2316	0.2042
HGAT	0.2372	0.2033	0.1783	0.2361	0.2104	0.1734
HRRPE	0.4369	0.3974	0.3728	0.4108	0.3845	0.3572

presents the superiority of the proposed HRRPE method compared to other models, where its precision, recall, and NDCG improvements for $L = 10$ compared to the second-best model (HGAT) ae approximately 24.81%, and 18.18%, and 17.32%, respectively.

In addition to the accuracy-based metrics, we also analyzed the performance of different RSs on the cold start users based on the health-score metric and report the relevant results in Table 9 for both the Allrecipes.com and Food.com datasets. The values of the health-score metric on the Allrecipes.com dataset obtained by HRRPE are 0.4369, 0.3974, and 0.3728 for $L = 10$, $L = 15$, and $L = 20$, respectively. These health-score values were the best results. The HPRR model was the second-best performer, and it achieved health-score values of 0.3651, 0.3362, and 0.3041 for recommendation list lengths $L = 10$, $L = 15$, and $L = 20$, respectively. From these results, we can infer that the food recommendations provided by the proposed HRRPE method were approximately 19.66%, 18.2%, and 22.59% healthier than those provided by the second-best model for $L = 10$, $L = 15$, and $L = 20$, respectively. The results obtained for the Food.com dataset also demonstrated that HRRPE outperforms other models in all recommendations list lengths. Therefore, HRRPE can not only alleviate the data sparsity problem, but it also is very effective in producing healthy and accurate food recommendations for cold start users. This achievement is mainly attributed to the proposed rating profile expansion technique, which significantly enhances the ability of food RSs to make food recommendations accurately even when the RSs face data sparsity and cold start problems.

4.6 Ablation study

The proposed method makes three main contributions: developing a rating profile expansion technique, a user community detection model, and a health-aware reliability measure. An important analysis is evaluating the importance of each contribution in enhancing the quality of food recommendations produced by the proposed method. Accordingly, an ablation study was performed in this section to demonstrate how the performance of the proposed method changes by ignoring each contribution. In particular, three scenarios are defined as follows:

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Scenario 1: In this scenario, we ignore the rating profile expansion technique in the structure of the proposed method. Thus, this scenario is helpful to determine the effect of the rating profile expansion technique on the performance of the proposed recommendation model.
Scenario 2: This scenario involves ignoring the proposed user community detection model in the recommendation procedure. In this case, the classic K-nearest neighbors approach is used instead of the proposed community detection model to predict virtual ratings. In this scenario, the effectiveness of the proposed user community detection model in improving the recommendation quality can be evaluated.
Scenario 3: This scenario aims to determine the importance of the proposed health-aware reliability measure in producing accurate and healthy food recommendations. To this end, we ignore this reliability measure in the recommendation process. In this scenario, the predicted virtual ratings are chosen based on their ratings rather than their reliability values for utilization in the rating profile expansion technique.

For the above-mentioned scenarios, some experiments were conducted on the Allrecipes.com and Food.com datasets in terms of different evaluation metrics for all users and cold start users. Figures 7 and 8 illustrate the results of the ablation study performed for all users and cold start users, respectively. Noticeably, the lowest performance was obtained in scenario 1, which ignored the proposed rating profile expansion technique in the recommendation process. Thus, this technique plays a crucial role in the proposed food recommendation system and significantly influences the quality of recommendations, considering both accuracy and health score. These results are expected because ignoring the rating profile expansion technique makes use of the original sparse user-food interaction matrix instead of the expanded matrix in the recommendation process. In addition, the efficiency of the developed food RS based on scenario 2 is better than scenario 3 in terms of all evaluation metrics and all datasets for all users and cold start users. As a result, it can

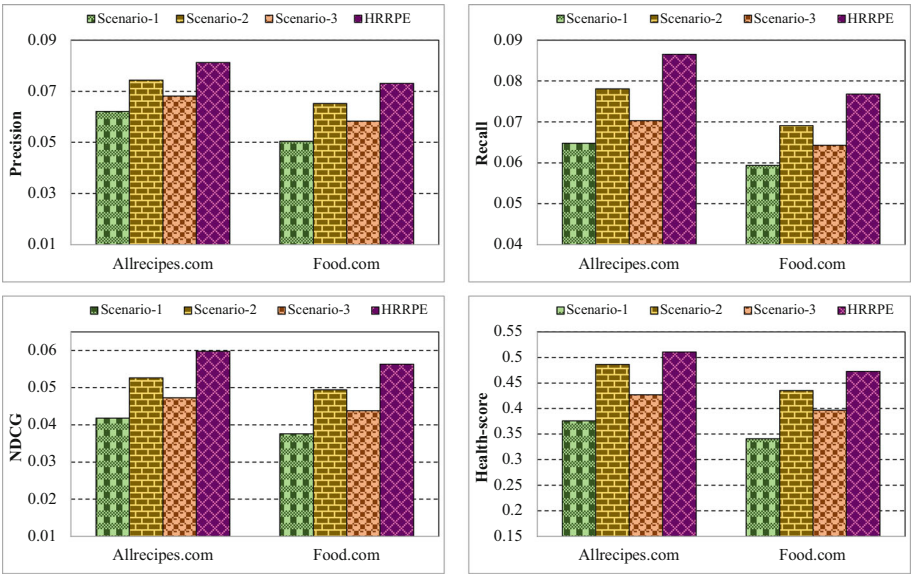


Fig. 7 Comparison between the proposed HRRPE method and different scenarios based on precision, recall, NDCG, and health-score metrics for the Allrecipes.com and Food.com datasets, all users, and $L = 10$

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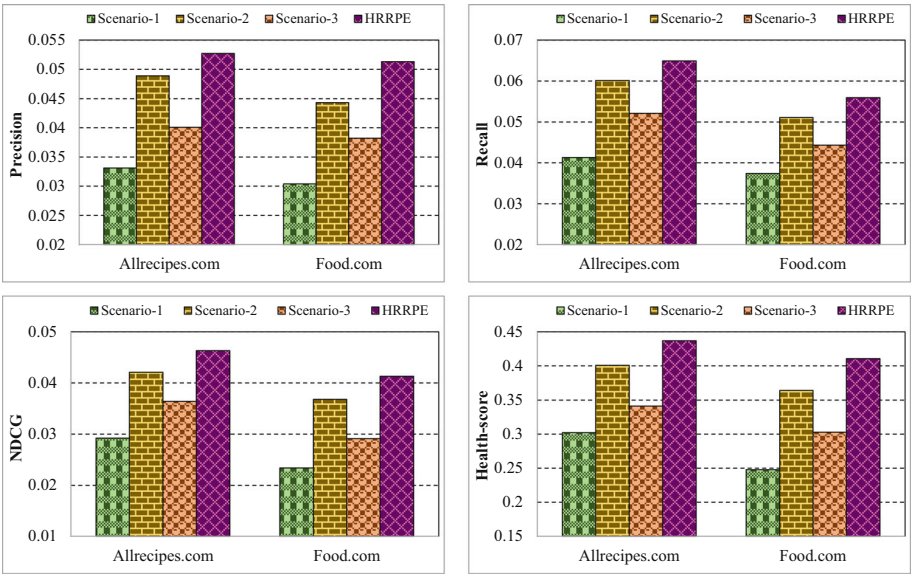


Fig. 8 Comparison between the proposed HRRPE method and different scenarios based on precision, recall, NDCG, and health-score metrics for the Allrecipes.com and Food.com datasets, cold start users, and $L = 10$

be inferred that the importance of the developed reliability score is higher than that of the developed user community approach in enhancing the effectiveness of the proposed RS, as the performance of the proposed method declined more here than in the scenario in which we ignore the community detection model when the recommendation process does not consider the reliability measure. The performance of the proposed HRRPE method was better than all three scenarios in both datasets and across all evaluation metrics for all users and cold start users. Accordingly, the recommendation quality will be enhanced when we consider all the contributions of the proposed method together in the recommendation process.

4.7 Statistical analysis

Statistical analysis is a theoretical approach for investigating the achieved results and demonstrating whether they are statistically significant. One of the well-known approaches to performing a statistical analysis is the Friedman test [61], which provides a nonparametric test mechanism to measure statistical differences between the results achieved by the models considered in the experiments. To this end, the models should be sorted descending based on the values of each evaluation metric separately. The model that obtains the best performance gets the first rank, whereas the last rank will be assigned to the worst model. Let t and q denote the number of models and datasets employed in the experiments, respectively. After performing the ranking strategy, the following equations are used to conduct the Friedman test:

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Table 10 Friedman test results for various evaluation metrics

Metrics	χ_F^2	F_F	$F(7,7)$	Significance
Precision	13.68	42.75	3.79	+
Recall	13.40	22.33	3.79	+
NDCG	13.45	24.45	3.79	+
Health score	13.86	99.00	3.79	+

$$F_F = \frac{(q-1)\chi_F^2}{q(t-1)-\chi_F^2} \tag{35}$$

where,

$$\chi_F^2 = \frac{12q}{t(t+1)}\left[\sum_{j=1}^{t} R_j^2 - \frac{t(t+1)^2}{4}\right] \tag{36}$$

and R_j denotes the average of ranks obtained by the j th model on different datasets. Notably, we set $t = 8$ and $q = 2$ for the Friedman test because eight recommendation models and two datasets were utilized in the experiments. Furthermore, a significance level of $\alpha = 0.05$ and degrees of freedom $F((t-1),(t-1)(q-1)) = F(7,7) = 3.79$ were established for the Fisher distribution to carry out the Friedman test. In this analysis, the null hypothesis was defined as all models performing equally. To determine whether this null hypothesis is accepted, we compared the value of F_F to 3.79: if F_F is less than 3.79, then the hypothesis is accepted, otherwise; it is rejected. For each evaluation metric, the Friedman test is separately conducted and the obtained results are reported in Table 10. These results indicate that all calculated values of F_F for various metrics exceeded 3.79. Consequently, the defined null hypothesis is rejected, confirming that the experimental results hold statistical significance.

5 Conclusions

Recently, food RSs have emerged in online food services to provide intelligent techniques to make personalized food suggestions for users. In this study, we introduce a novel food RS based on an effective rating profile expansion technique. Initially, we formulated a probabilistic model to assess the rating profiles of users, determining if each rating profile encompasses adequate ratings to predict precise food recommendations. Then, the proposed rating profile expansion technique is performed on each profile that suffers from insufficient ratings to create more effective profiles, leading to a user–food interaction matrix that is denser than the original one. To this end, some virtual ratings are predicted based on the preferences of users who are grouped in the community of the target user using a community detection model. We evaluated our model on two real-world datasets and proved its superiority over other methods in terms of three performance measures and one health-based measure. The proposed rating profile expansion technique can alleviate the data sparsity problem by incorporating virtual ratings that are healthy and accurate into the rating profiles. This is achieved by evaluating the virtual ratings using a health-aware reliability measure that considers both accuracy and health criteria simultaneously. By selecting the most reliable virtual ratings,

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we can guarantee that the incorporated virtual ratings are healthy and accurate, thus significantly improving food recommendations. Another advantage of the developed system is that it automatically determines the minimum number of ratings required for each user to make a reliable recommendation. This helps in calculating the exact number of ratings to be added to those profiles whose available ratings are insufficient. Although the proposed method provides significant advantages, it does not utilize important data resources, such as food ingredients and the social relationships of the users. These data resources may have a considerable impact on the recommendation performance. Moreover, user characteristics, such as age, sex, and medical history, are important pieces of information that can be utilized in the proposed method to make healthier food suggestion using the obtainable information for each user. These shortcomings can be addressed in future work.

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Authors' contributions SA and MR conceived of the presented idea. The writing is done by SA, MR, and SMJJ. SA developed the theory and carried out the experiments. MR and SMJJ verified the analytical methods. MO and VF supervised the project. All authors have read and approved the final manuscript.

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Availability of data and material The datasets used in this study can be available on reasonable requests.

Declarations

Conflict of interest The authors declare no competing interests.

Ethics approval and consent to participate The data used in this paper are without ethical experiments.

Consent for publication The authors involved in this paper agree to publish this paper.

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References

1. Shao B, Li X, Bian G (2021) A survey of research hotspots and frontier trends of recommendation systems from the perspective of knowledge graph. Expert Syst Appl 165:113764
2. Lee J-w, Choi M, Sael L, Shim H, Lee J (2022) Knowledge distillation meets recommendation: collaborative distillation for top-N recommendation. Knowl Inf Syst 64(5):1323–1348
3. Yengikand AK, Meghdadi M, Ahmadian S, Jalali SMJ, Khosravi A, Nahavandi S (2021) Deep Representation Learning using Multilayer Perceptron and Stacked Autoencoder for Recommendation Systems. 2021 IEEE International Conference on Systems, Man, and Cybernetics (SMC), pp 2485–2491
4. Ahmadian M, Ahmadi M, Ahmadian S, Jalali SMJ, Khosravi A, Nahavandi S (2021) Integration of Deep Sparse Autoencoder and Particle Swarm Optimization to Develop a Recommender System. 2021 IEEE International Conference on Systems, Man, and Cybernetics (SMC), pp 2524–2530

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5. Davtalab M, Alesheikh AA (2021) A POI recommendation approach integrating social spatio-temporal information into probabilistic matrix factorization. Knowl Inf Syst 63(1):65–85
6. Liu H, Zheng C, Li D, Zhang Z, Lin K, Shen X et al (2022) Multi-perspective social recommendation method with graph representation learning. Neurocomputing 468:469–481. https://doi.org/10.1016/j.neucom.2021.10.050
7. Liu H, Zheng C, Li D, Shen X, Lin K, Wang J et al (2022) EDMF: efficient deep matrix factorization with review feature learning for industrial recommender system. IEEE Trans Industr Inf 18(7):4361–4371. https://doi.org/10.1109/TII.2021.3128240
8. Bellogín A, Castells P, Cantador I (2017) Statistical biases in information retrieval metrics for recommender systems. Inf Retrieval J 20(6):606–634
9. Rahayu NW, Ferdiana R, Kusumawardani SS (2022) A systematic review of ontology use in E-Learning recommender system. Comput Educ: Artif Intell 3:100047
10. Mao M, Chen S, Zhang F, Han J, Xiao Q (2021) Hybrid ecommerce recommendation model incorporating product taxonomy and folksonomy. Knowl-Based Syst 214:106720
11. Ahmadian S, Meghdadi M, Afsharchi M (2018) A social recommendation method based on an adaptive neighbor selection mechanism. Inf Process Manage 54(4):707–725
12. Liu Z, Wang X, Ma Y, Yang X (2022) Relational metric learning with high-order neighborhood interactions for social recommendation. Knowl Inf Syst 64(6):1525–1547
13. Ahmadian S, Ahmadian M, Jalili M (2022) A deep learning based trust- and tag-aware recommender system. Neurocomputing 488:557–571. https://doi.org/10.1016/j.neucom.2021.11.064
14. Ahmadian S, Joorabloo N, Jalili M, Meghdadi M, Afsharchi M, Ren Y (2018) A Temporal Clustering Approach for Social Recommender Systems. 2018 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM), pp 1139–1144
15. Rezaeimehr F, Moradi P, Ahmadian S, Qader NN, Jalili M (2018) TCARS: time- and community-aware recommendation system. Futur Gener Comput Syst 78:419–429. https://doi.org/10.1016/j.future.2017.04.003
16. Yengikand AK, Meghdadi M, Ahmadian S (2023) DHSIRS: a novel deep hybrid side information-based recommender system. Multimedia Tools Appl 82(22):34513–34539. https://doi.org/10.1007/s11042-023-15021-9
17. Nam LNH (2022) Towards comprehensive approaches for the rating prediction phase in memory-based collaborative filtering recommender systems. Inf Sci 589:878–910
18. Lian J, Zhang F, Xie X, Sun G (2017) CCCFNet: A Content-Boosted Collaborative Filtering Neural Network for Cross Domain Recommender Systems. Proceedings of the 26th International Conference on World Wide Web Companion. Perth, Australia: International World Wide Web Conferences Steering Committee, pp 817–818
19. Zamanzadeh Darban Z, Valipour MH (2022) GHRS: graph-based hybrid recommendation system with application to movie recommendation. Expert Syst Appl 200:116850
20. Fransen HP, Boer J, Beulens JW, De Wit GA, Bueno-de-Mesquita HB, Hoekstra J et al (2017) Associations between lifestyle factors and an unhealthy diet. Eur J Pub Health 27(2):274–278
21. Akdeniz Kudubes A, Ayar D, Bektas İ, Bektas M (2022) Predicting the effect of healthy lifestyle belief on attitude toward nutrition, exercise, physical activity, and weight-related self-efficacy in Turkish adolescents. Arch Pediatr 29(1):44–50
22. Rostami M, Oussalah M, Farrahi V (2022) A novel time-aware food recommender-system based on deep learning and graph clustering. IEEE Access 10:52508–52524
23. Rostami M, Berahmand K, Forouzandeh S, Ahmadian S, Farrahi V, Oussalah M (2024) A novel healthy food recommendation to user groups based on a deep social community detection approach. Neurocomputing 576:127326. https://doi.org/10.1016/j.neucom.2024.127326
24. Rostami M, Farrahi V, Ahmadian S, Mohammad Jafar Jalali S, Oussalah M (2023) A novel healthy and time-aware food recommender system using attributed community detection. Expert Syst Appl 221:119719. https://doi.org/10.1016/j.eswa.2023.119719
25. Ahmadian S, Rostami M, Jalali SMJ, Oussalah M, Farrahi V (2022) Healthy food recommendation using a time-aware community detection approach and reliability measurement. Int J Comput Intell Syst 15(1):105. https://doi.org/10.1007/s44196-022-00168-4
26. Majumder BP, Li S, Ni J, Julian M (2019) Generating personalized recipes from historical user preferences. Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), pp 5976–5982
27. Musto C, Starke A, Trattner C, Rapp A, Semeraro G (2021) Exploring the effects of natural language justifications in food recommender systems. Proceedings of the 29th ACM Conference on User Modeling, Adaptation and Personalization: ACM, pp 147–57

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28. Musto C, Trattner C, Starke A, Semeraro G (2020) Towards a knowledge-aware food recommender system exploiting holistic user models. Proceedings of the 28th ACM Conference on User Modeling, Adaptation and Personalization: ACM, pp 333–337
29. Ge M, Elahi M, Fernaández-Tobías I, Ricci F, Massimo D (2015) Using tags and latent factors in a food recommender system. Proceedings of the 5th International Conference on Digital Health: ACM, pp 105–112
30. Singh M (2020) Scalability and sparsity issues in recommender datasets: a survey. Knowl Inf Syst 62(1):1–43
31. Ahmadian S, Joorabloo N, Jalili M, Ahmadian M (2022) Alleviating data sparsity problem in time-aware recommender systems using a reliable rating profile enrichment approach. Expert Syst Appl 187:115849. https://doi.org/10.1016/j.eswa.2021.115849
32. Ahmadian S, Afsharchi M, Meghdadi M (2019) A novel approach based on multi-view reliability measures to alleviate data sparsity in recommender systems. Multimedia Tools Appl 78(13):17763–17798. https://doi.org/10.1007/s11042-018-7079-x
33. Natarajan S, Vairavasundaram S, Natarajan S, Gandomi AH (2020) Resolving data sparsity and cold start problem in collaborative filtering recommender system using Linked Open Data. Expert Syst Appl 149:113248
34. Tahmasebi F, Meghdadi M, Ahmadian S, Valiallahi K (2021) A hybrid recommendation system based on profile expansion technique to alleviate cold start problem. Multimedia Tools Appl 80(2):2339–2354
35. Formoso V, Fernández D, Cacheda F, Carneiro V (2013) Using profile expansion techniques to alleviate the new user problem. Inf Process Manage 49(3):659–672
36. Ahmadian S, Afsharchi M, Meghdadi M (2019) An effective social recommendation method based on user reputation model and rating profile enhancement. J Inf Sci 45(5):607–642
37. Ahmadian M, Ahmadi M, Ahmadian S (2022) A reliable deep representation learning to improve trust-aware recommendation systems. Expert Syst Appl 197:116697. https://doi.org/10.1016/j.eswa.2022.116697
38. Denson W (1998) The history of reliability prediction. IEEE Trans Reliab 47(3):SP321–SP328
39. Ahmadian M, Ahmadian S, Ahmadi M (2023) RDERL: Reliable deep ensemble reinforcement learning-based recommender system. Knowl-Based Syst 263:110289. https://doi.org/10.1016/j.knosys.2023.110289
40. Bobadilla J, Gutiérrez A, Ortega F, Zhu B (2018) Reliability quality measures for recommender systems. Inf Sci 442–443:145–157. https://doi.org/10.1016/j.ins.2018.02.030
41. Hernando A, Bobadilla J, Ortega F, Tejedor J (2013) Incorporating reliability measurements into the predictions of a recommender system. Inf Sci 218:1–16. https://doi.org/10.1016/j.ins.2012.06.027
42. Toledo RY, Alzahrani A, Martínez L (2019) A food recommender system considering nutritional information and user preferences. IEEE Access 7:96695–96711
43. Gao X, Feng F, He X, Huang H, Guan X, Feng C et al (2020) Hierarchical attention network for visually-aware food recommendation. IEEE Trans Multimedia 22(6):1647–1659
44. Gao X, Feng F, Huang H, Mao X-L, Lan T, Chi Z (2022) Food recommendation with graph convolutional network. Inf Sci 584:170–183
45. Mathur A, Juguru SK, Eirinaki M (2019) A graph-based recommender system for food products. 2019 First International Conference on Graph Computing (GC). IEEE, Laguna Hills, pp 83–87
46. Trattner C, Elsweiler D (2017) Investigating the healthiness of internet-sourced recipes: Implications for meal planning and recommender systems. Proceedings of the 26th International Conference on World Wide Web. ACM, pp 489–498
47. Wang W, Duan L-Y, Jiang H, Jing P, Song X, Nie L (2021) Market2Dish: Health-aware food recommendation. ACM Trans Multimed Comput Commun Appl 17(1):1–19
48. Singh PK, Pramanik PKD, Choudhury P (2022) Mitigating sparsity using Bhattacharyya Coefficient and items' categorical attributes: improving the performance of collaborative filtering based recommendation systems. Appl Intell 52(5):5513–5536. https://doi.org/10.1007/s10489-021-02462-8
49. Kolahkaj M, Harounabadi A, Nikravanshalmani A, Chinipardaz R (2021) Incorporating multidimensional information into dynamic recommendation process to cope with cold start and data sparsity problems. J Ambient Intell Humaniz Comput 12(10):9535–9554. https://doi.org/10.1007/s12652-020-02695-4
50. Shen R-P, Zhang H-R, Yu H, Min F (2019) Sentiment based matrix factorization with reliability for recommendation. Expert Syst Appl 135:249–258. https://doi.org/10.1016/j.eswa.2019.06.001
51. Xie H, Lui JCS (2015) Mathematical Modeling and Analysis of Product Rating with Partial Information. ACM Trans Knowl Discov Data 9(4):1–33
52. Bishop CM (2006) Pattern recognition and machine learning. Springer-Verlag, New York
53. Matoussek J, Vondrak J (2001) The Probabilistic Method. Charles University, Prague

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54. Das K, Samanta S, Pal M (2018) Study on centrality measures in social networks: a survey. Soc Netw Anal Min 8(1):13. https://doi.org/10.1007/s13278-018-0493-2
55. Gao R, Li S, Shi X, Liang Y, Xu D (2021) Overlapping community detection based on membership degree propagation. Entropy 23(1):15
56. Cui Y, Li X, Li J, Wang H, Chen X (2022) A survey of sampling method for social media embeddedness relationship. ACM Comput Surv. https://doi.org/10.1145/3524105
57. Pecune F, Callebert L, Marsella S (2020) A recommender system for healthy and personalized recipes recommendations. HealthRecSys@ RecSys, pp 15–20
58. Morol K, Rokon SJ, Hasan IB, Saif AM, Khan RH, Das SS (2020) Food recipe recommendation based on ingredients detection using deep learning. International Conference on Computing Advancements (ICCA 2022). ACM, pp 1–7
59. Chavan P, Thoms B, Isaacs J (2021) A Recommender System for Healthy Food Choices: Building a Hybrid Model for Recipe Recommendations using Big Data Sets. Proceedings of the 54th Hawaii International Conference on System Sciences, pp 3774–83
60. Tian Y, Zhang C, Metoyer R, Chawla NV (2022) Recipe recommendation with hierarchical graph attention network. Front Big Data 4:1–13. https://doi.org/10.3389/fdata.2021.778417
61. Friedman M (1937) The use of ranks to avoid the assumption of normality implicit in the analysis of variance. J Am Stat Assoc 32(200):675–701

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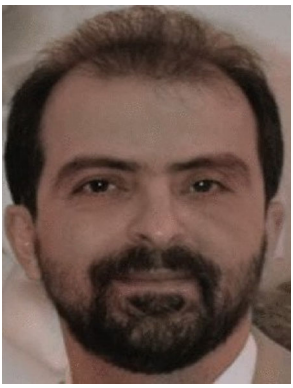


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